



EDI ENVIRONMENTAL DYNAMICS INC.
Natural Resource Consultants

**Salmon Planning within the
White River First Nation
Traditional Territory
CRE-053N-03**

Prepared for:

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and

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General Delivery
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DISCLAIMER

Although the information presented in this report is designed to act as a comprehensive compilation of existing information, findings are based on pre-existing literature. The information presented is only a summary of the results of each study, and it is therefore recommended that users of this information verify specific details before it is used to base future management decisions.

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1.0 INTRODUCTION

Salmon, particularly chinook (*Oncorhynchus tshawytscha*), are of great interest to the White River First Nation (WRFN). This project seeks to obtain a more detailed understanding of known chinook distributions within the entire WRFN Traditional Territory by seeking out such existing documented information and compiling it in a single document. To achieve this, a local researcher was hired to gather information about the fisheries resource within the WRFN Traditional Territory. A community workshop/open house was then held where community members and WRFN representatives discussed the previously gathered information and provided further Traditional and local fisheries knowledge. Fisheries data gathered in these exercises has been summarized in this report.

1.1 Background

In 2001, the WRFN undertook a fisheries inventory and assessment in the Snag Creek watershed, which helped increase public awareness of salmon in the community (Tobler et al. 2001). The results of this study included a renewed interest in determining the extent of chinook utilization in the upper White River. In the past three years the WRFN has conducted two additional projects in this regard. These projects were entitled ‘2001 Fish and Fish Habitat Inventory and Assessment of Snag Creek Watershed’ and ‘2002 Upper White River Chinook Reconnaissance’. While no chinook were documented in these studies, there was some evidence of past use of the area. This work also generated local discussion about possible chinook stock-restoration opportunities in the area. Interest has also been expressed in other salmon projects as well. Some examples include:

- Surveys of known and potential chum (*Oncorhynchus keta*) runs utilizing the Koidern River, Snag River, and Sanpete Creek;
- Investigations of salmon usage within the Upper Tanana River drainage, with a focus on Scottie Creek;
- Possible “reintroduction” of chinook salmon in Snag Creek.

While it is certain that a local interest in salmon is developing, there has been a lack of overall salmon management planning for the WRFN Traditional Territory. Further in this regard, a comprehensive understanding of the salmon resource in the Traditional Territory has not been available. Before further salmon related work is conducted in the area, there has been a need to develop a better understanding of the salmon resource. This project will ensure that stakeholders and First Nations have access to an accessible and comprehensive documentation of existing salmon knowledge, as well as some data gaps and potential issues within watersheds located in the vicinity of the WRFN Traditional Territory. From this basis, further salmon related projects can be planned and an overall vision for the future of salmon related activities in the area can be developed.

A variety of current and potential fish habitat issues may exist throughout the WRFN Traditional Territory. The area has seen a myriad of historic and current development activities that may have impacted upon fish habitat utilisation. As well the impacts of climate change are having, and will continue to have, significant impacts upon the area. Construction of the Alaska Highway, construction/use/abandonment of the old pipeline found roughly along the route of the Alaska Highway, current and past mineral exploration and extraction (including placer), and extensive past timber harvesting may have impacted streams throughout the study area. While some information regarding particular issues was collected in this study, such issues were not the primary focus of the project. Therefore the fish habitat issues documented in this report represent only a small number of those that may exist.

1.2 Study Area

The WRFN Traditional Territory is located within the southwest portion of Yukon Territory (Figure 1). WRFN members have significant social and cultural ties with members of the Kluane First Nation and with Native peoples of the upper Tanana watershed. As well, a 100% overlap in Traditional Territory exists between the WRFN and the Kluane First Nation.

For organizational and mapping purposes, the WRFN Traditional Territory (and additional surrounding areas) were separated into the following watersheds:

1. Upper White River watershed (upstream of the Donjek River confluence)
2. Donjek River watershed (not including Kluane River Watershed)
3. Kluane River watershed
4. Upper Tanana River watershed (upstream of the Tok River confluence)

2.0 METHODS

The project was conducted in the following phases:

- 1) Information gathering and analysis;
- 2) Digital mapping of collected information;
- 3) Workshop and open house;
- 4) Reporting/Data compilation.

2.1 Information Gathering and Analysis

Information related to fish distributions, obstructions to fish migration and fisheries issues within the WRFN Traditional Territory were collected and compiled.

A WRFN researcher was hired to initiate the data collection process, and conducted research in both Alaska and the Yukon (over a period of twelve days). Yukon fisheries data sources consulted included the Fisheries Inventory Summary System (FISS), Department of Fisheries and Oceans Library and Stream Files, consultant reports, Yukon College Library, Yukon Energy, Mines and Resources Library (formerly DIAND library), the Yukon Territory Government Renewable Resources Library, and information on file with the WRFN. Consultations in Alaska were with the U.S. Fish and Wildlife, including managers of the Tetlin National Wildlife Refuge, and the Alaska Department of Fish and Game.

2.2 Digital Mapping

Digital mapping was performed by EDI staff using MapInfo. Large-scale maps were first produced showing fish distributions. These rough maps were used as visual aids during the workshop and open house (see Section 2.3). They were then modified to reflect changes and additions provided by workshop participants, as well as further information that was subsequently gathered. These maps are presented in Appendix A. All GIS points were referenced with codes corresponding to the date and source of the information, which is summarized in Appendix B.

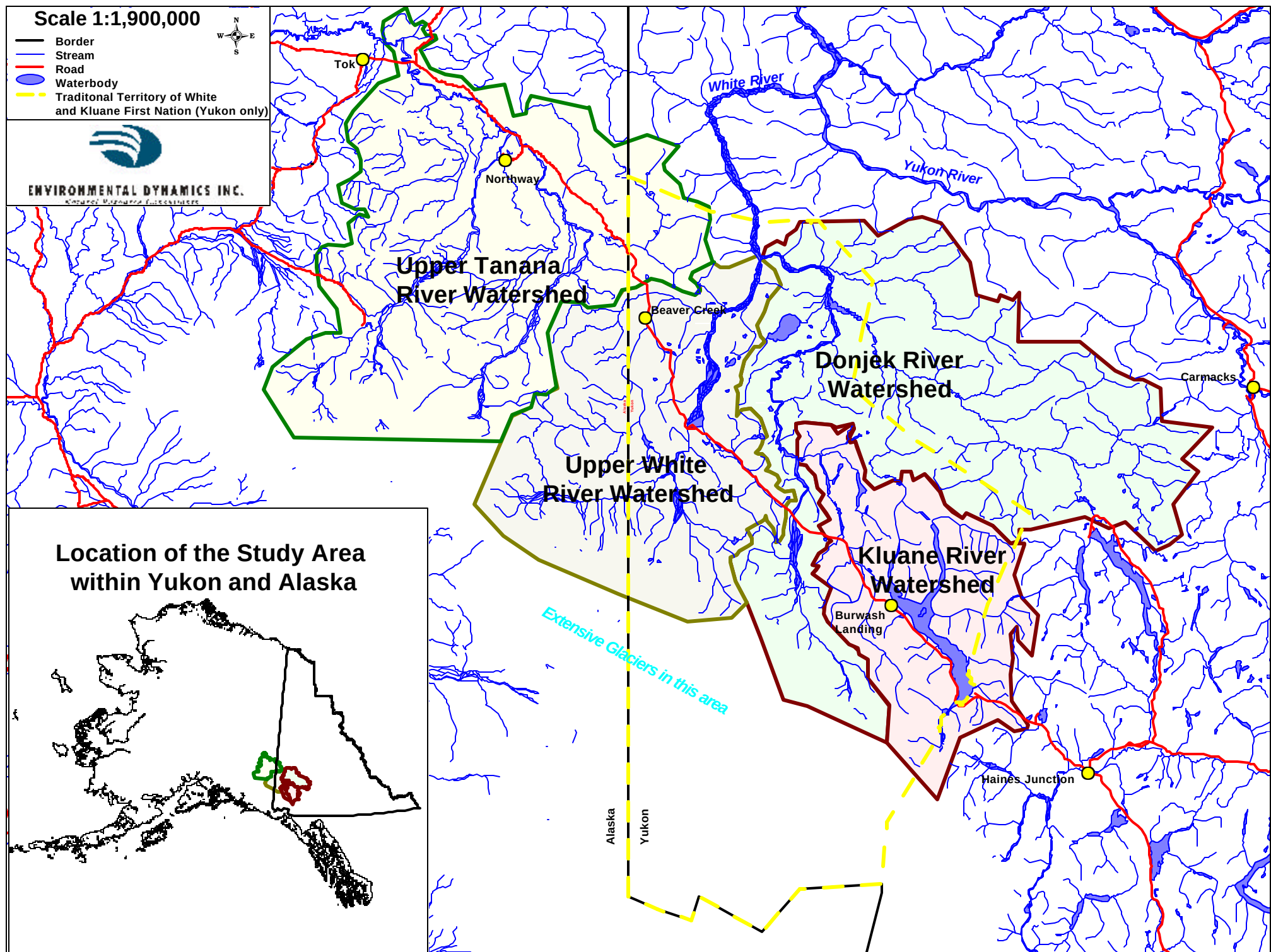


Figure 1. Study Area Map.

2.3 Workshop/Open House

On May 27th and 28th 2004, a workshop and open house was held at the community centre in Beaver Creek (see Figures 2 and 3). Representatives from DFO, the Yukon Salmon Committee, the Yukon River Panel, and the Yukon Fish and Wildlife Management Board attended and provided a variety of information and input in regards to salmon management in the Yukon. Representatives of the WRFN as well as community members and elders also participated (see Appendix C for participant list). A representative of the United States Fish and Wildlife Service from the Tetlin Wildlife Preserve (Connie Friend) attended along with Tetlin Elder Ellen Klemmit.

The theme and structure of the workshop placed emphasis on encouraging further involvement of the WRFN and local residents in the salmon management process within the Yukon. Considerable interest was expressed by participants not only in becoming more involved in such management, but also in seeing further salmon related projects undertaken at the local level. Several possible project ideas were enthusiastically discussed.

As well, a particularly successful component of the workshop was a thorough discussion of local and Traditional Knowledge with regard to salmon and other species of fish in the White and Tanana River watersheds. Participants shared their knowledge and experiences, which was recorded for reference in this report. Workshop participants were informed at the time that all information provided by them during the workshop could be documented in this report, thereby becoming public information. Information from this workshop is cited in this report as “Salmon Workshop, 2004.”



Figure 2. Participants in the salmon information workshop, Beaver Creek 2004



Figure 3. Traditional net for freshwater fish (from the Scottie Creek area) and other artifacts on display at salmon information workshop, Beaver Creek 2004—courtesy of David Johnny.

2.4 Reporting/Data Compilation

All relevant documented fisheries information found during the compilation of this report was summarized by EDI biologists and put into table format including known locations of salmon and freshwater fish species (Appendix B).

The following sections summarize the information collected for each of the major watersheds identified in the WRFN Traditional Territory.

For reference, Traditional Knowledge and community ideas documented at the community workshop have been identified independently as such. This knowledge and input has also been incorporated into the fabric of this report. It should be noted that while Traditional Knowledge gathered in the workshop is identified separately, other sources of Traditional Knowledge have also been synthesized into this report.

3.0 RESULTS/DISCUSSION

The four study watersheds (Donjek, upper White, Kluane and upper Tanana) are home to sixteen species of fish including three species of salmon, nine species considered sport fish, and five species of non-sport fish (see Table 1 below).

Table 1. Fish species documented in the WRFN First Nation Traditional Territory and surrounding areas

Category	Common Name	Scientific Name	Species Code
Salmon species	chinook salmon	<i>Oncorhynchus tshawytscha</i>	CH
	chum salmon	<i>Oncorhynchus keta</i>	CM
	coho salmon	<i>Oncorhynchus kisutch</i>	CO (Tanana only)
Freshwater¹ game fish species	dolly varden	<i>Salvelinus malma</i>	DV (Tanana only)
	arctic grayling	<i>Thymallus arcticus</i>	GR
	least cisco	<i>Coregonus sardinella</i>	CS
	broad whitefish	<i>Coregonus nasus</i>	BW
	lake whitefish	<i>Coregonus clupeaformis</i>	LW
	burbot	<i>Lota lota</i>	BB
	inconnu	<i>Stenodus leucichthys</i>	IN
	northern pike	<i>Esox lucius</i>	NP
Other fish species	lake trout	<i>Salvelinus namaycush</i>	LT
	longnose sucker	<i>Catostomus catostomus</i>	LSU
	lake chub	<i>Couesius plumbeus</i>	LKC
	slimy sculpin	<i>Cottus cognatus</i>	CCG
	round whitefish	<i>Prosopium cylindraceum</i>	RW
	arctic lamprey	<i>Lampetra japonica</i>	AL

¹ As categorized by Yukon Territorial Government freshwater fishery regulations.

It should be noted that the issues identified herein regarding of obstructions and constraints to fish habitat utilization constitute only a limited account. While some such issues were identified through the course of the research, to generate a complete account of such items was not an objective of this project. As a result, the identified data gaps and potential projects do not address all obstructions/constraints that may be impacting upon fish habitat utilisations.

3.1 Upper White River Watershed (upstream of Donjek River)

The headwaters of the White River originate from the northern slopes of the eastern portion of the Wrangell Mountains in Alaska. All the headwaters within Alaska are located within the Wrangell-St. Elias National Park and the Wrangell-St. Elias National Preserve. The headwaters of the White River in the Yukon include the Klutlan Glacier located in the Kluane Wildlife Sanctuary. The only community currently occupied within the study area is Beaver Creek, home of the White River First Nation and the last community on the Canadian portion of the Alaska Highway, near the Canada/U.S. Border.

Major tributaries to the White River within the study area include Lake Creek, Snag Creek, Dry Creek, Sanpete Creek and the Koidern River. Twelve species of fish have been documented within the major streams that make up this portion of the White River watershed (see Table 2).

Table 2: Fish species documented in the upper White watershed

Category	Common Name
Salmon species	Chum salmon, chinook salmon
Freshwater game fish species	Arctic grayling, lake trout, lake whitefish, burbot, northern pike, least cisco, whitefish general
Other fish species	lake chub, slimy sculpin, longnose sucker, round whitefish

3.1.1 Fish Distribution and Habitat

A moderate level of fish diversity was found within this watershed, with available information indicating some utilization by chum salmon and sporadic utilization by chinook salmon (see Table 3). Little information is available on portions of the watershed, however, recent research conducted in 1998, 2001 & 2002 by the White River First Nation has provided some further insight into some of the more inaccessible areas of the upper White River drainage (Otto, 1998, Tobler 2001 and 2003). In the 1970s considerable research was conducted on specific components of the watershed in association with the proposed Alaska Highway pipeline route. However, other, often inaccessible, areas of the watershed have had little to no scientific/technical research conducted on them to assess fisheries utilization/habitat of such streams. Table 3 lists all streams within the upper White River watershed with documented species of fish for each. For reference, all streams/species identified in the community workshop have been also listed in a separate column.

The information gathered has indicated that streams within the upper White River watershed host either a moderate or low diversity of fish species. Streams containing a moderate diversity of fish species included the upper White River itself, Snag Creek and tributaries, as well as the Koidern River.

In general, salmon habitat in this area appears to be under-utilized. Only the Koidern River was identified as having a consistently documented run of chum salmon. The returns of chum salmon documented there have appeared variable in number (Horler, 1994). Many areas with groundwater sources have been identified in the Koidern, indicating potentially good habitat for chum salmon spawning. As well, in 1997 the lower portion of Beaver Creek (below Enger Creek) was identified as having groundwater sources (Otto, 1997). Limited research has found evidence of very small numbers of spawning chum has also been observed in a slough within the White River floodplain, 3 kilometres upstream of Snag Creek (Johnston, 1992). In 1984, a radio tagged chinook was located in the White River mainstem, approximately 5km downstream from Snag Creek. This should be considered a significant finding, possibly indicating a possible undocumented small, established run of chinook salmon within the upper White River (Milligan et al, 1984). In 2002, a study undertaken by the WRFN identified what appeared to be favourable chinook habitat in Snag and Sanpete Creeks, and, Traditional Knowledge documented in the same report indicated that chinook had been observed in Sanpete (Tobler, 2003). It is also interesting to note that in 1995 chinook salmon were observed on multiple occasions in Enger Creek, a tributary of Beaver Creek. Currently, Enger Creek does not appear to be good chinook habitat and is clogged with beaver dams. Further, in 1996, minnow trapping conducted by Fisheries and Oceans in Enger creek found no juvenile chinook (Salmon Workshop, 2004).

Table 3: Streams¹ within the upper White watershed with documented fish species

<u>Stream Name</u>	<u>Identified by Community in Workshop³</u>	<u>Salmon³</u>	<u>Freshwater Species³</u>
O'Brien Creek— (Just below Donjek)			GR, RW, BB, CCG, LKC, LSU
White River Mainstem (above Donjek)		CH ² , CM ²	GR, LW, RW, BB, LSU, CCG, LKC, NP
Ten Mile Creek			CCG
Lake Creek			GR, CCG, LSU, LKC
Fish Hole Lake (Lake Creek)			GR, NP
Snag Creek	CH	CH	GR, RW, CCG, LKC, BB, LSU
Beaver Creek (Snag)			GR, RW, CCG
Enger Creek (Beaver)	CH	CH ²	GR, CCG
Enger Lakes (Enger)			GR, NP, WF
Tchawsahmon Lake (Beaver)			LT, LW, BB, NP, GR, CS
Dry Creek	GR		GR
Sanpete Creek	GR		GR
Koidern River	CH, CM, WF, GR	CH ² , CM	GR, LW, RW, LSU, CCG, NP, BB, LKC
Pickhandle Lake (Koidern)	LW, LSU, NP, GR		WF, NP, GR, LSU
Wolf Creek (Koidern)	WF/LW		LW, CCG, GR
Long's Creek (Wolf)			LW, CCG, GR
Unnamed Creek— mile 37.8 (Koidern)			GR
Grafe Creek	GR, WF/LW		GR, BB, WF /LW
Lake Creek	GR		GR, LSU
Unnamed tributary to Lake Creek		CM (eggs) ²	
Rabbit Creek			
Generc River			
Boulder Creek			
Soda Creek			
Big Boundary Creek			
Little Boundary Creek			
Cache Creek			
Salt Lake			NP
Manh Ts'eeek			WF, GR, LSU, NP

¹Only streams with names appearing on 1:250,000 topographic maps are included.

²Reports of salmon in this location are isolated and not confirmed on a consistent basis.

³Refer to Table 1 for species codes.

3.1.2 Resource Use

Members of the White River First Nation have used this area extensively for generations. Some known locations of historic fish camps and settlements have been identified, with certain locations (particularly the village of Snag) likely to have been associated with the harvest of salmon (Tobler, 2003; Salmon Workshop, 2004). Also, streams and lakes with significant populations or migrations of freshwater fish have been, and in some cases still are, utilized extensively (Salmon Workshop, 2004). Recreational fishing occurs along the Alaska Highway, as well as in some of the larger lakes of the area. Table 4 lists some significant historic and current fisheries resource use in the area.

Table 4: Summary of fisheries resource use in the upper White watershed

Stream	Comments	Data Source
<i>Fish Hole Lake</i>	<i>Important Fishery for Beaver Creek People.</i>	<i>Dobrowolsky 1997</i>
<i>Snag Creek</i>	<i>An elder reported that chinook were caught with willow nets in Snag Creek near its mouth.</i> <i>Snag is a historic village site, likely a past hunting and fishing settlement.</i>	<i>Tobler, 2003</i> <i>Dobrowolsky 1997</i>
<i>Koidern River/Long's & Wolf Creeks</i>	<i>Lake "whitefish in this region were heavily exploited by domestic fishermen during the time when the fish were congregating in pools."</i> <i>Due to the high volumes of traffic along the Alaska Highway, it is likely that the Koidern River and tributaries are fished regularly.</i>	<i>Beak, 1977</i> <i>EDI</i>
<i>Tchawsahmon Lake</i>	<i>Old settlement on this lake. Accounts indicate an important subsistence fishery.</i> <i>A recreational fishery has existed more recently, with some possible continuation today.</i>	<i>Dobrowolsky 1997</i> <i>Penikett, 1993</i>

It is interesting to note the lack of available references to any fishery on the upper White River mainstem.

3.1.3 Obstructions/Constraints

The primary obstruction to fish passage identified in the upper White River watershed was related to Beaver activity in the Snag Creek system. This included Snag Creek itself, as well as Enger Creek. There was also a mention of a change in the flow of Enger Creek since construction of the Alaska Highway (Salmon Workshop, 2004).

The influence of climate change on fish habitat must be considered. Such influence may have current and future effects on a variety of environmental parameters that influence the availability and quality of fish habitat. Such influences may affect surface and subterranean hydrology, alter stream channels and substrates, and change the composition of riparian zone habitats. Such influences are complex with multiple factors directly and indirectly affecting each other and fish habitats.

Also, the question should be considered as to how the sediment load of the upper White River may act as a limiting factor in terms of salmon habitat (i.e. how it impacts migration).

3.1.4 Data Gaps and Potential Projects

Significant knowledge gaps have been identified and listed in Table 5.

Table 5: Significant data gaps within the upper White watershed. (as identified by authors).

• A limited amount of information regarding chum salmon populations in the Koidern River.
• Little is known about chinook habitat usage upstream of the Donjek River.
• A lack of information exists on historic activity of salmon in the watershed, particularly Snag Creek and its tributaries.
• A general lack of information regarding fish utilization of the upper White River system. Note that no information was collected in regard to the watershed upstream of the Alaska Highway.
• While a general understanding of the presence/absence of freshwater species exists, there is a significant absence of understanding in regards to the dynamics and abundance of these populations.
• Changes in stream channel habitat, including surface/subterranean hydrology, channel/substrate and riparian corridors that may have resulted from a variety of past and ongoing influences, such as climate change, forestry, construction of the Alaska Highway, and construction of the currently abandoned pipeline found roughly along the route of the Alaska Highway as it crosses the entire WRFN Traditional Territory.

Several ideas related to future work were identified in the community workshop and are listed below in Table 6.

Table 6: Fisheries related ideas for the upper White watershed identified in community workshop.

• Gather information that is meaningful for the White River First Nation.
• Ensure community involvement, including hiring and training, in projects.
• Study changes in River channels.
• Beaver and Beaver Dam removal (Enger/Beaver/Snag watershed).
• Try to reintroduce salmon (chinook) into Snag Creek.
• Study Dog Salmon in Koidern and compare with those from the Donjek to see changes.
• Nitrogen analysis of tree cores to scientifically establish the historic presence of salmon in Snag Creek.
• Conduct water temperature analysis of Snag Creek and compare with water temperatures from other creeks.
• Restart salmon in the classroom.
• Re-blaze historic trails.

Based on all reviewed information, including ideas/concerns/recommendations identified in the community workshop, some potential projects to be considered for the upper White River watershed are listed in Table 7 below:

Table 7: Potential fisheries projects for the Donjek watershed (as identified by authors).

• Identify suspected (historical versus present) changes in fish habitat within specific streams or portions thereof; and, conduct assessment of such changes in relation to suspected impacts, such as climate change, forestry, highway construction, and pipeline development.
• Assess feasibility of utilising marine derived nutrient analysis of tree cores as a methodology for assessing the historic presence/absence of salmon in the Snag/Beaver/Enger Creek and Sanpete watersheds.
• Completion of work previously begun to assess the feasibility of potential chinook stock restoration in Snag Creek.
• Assess feasibility of restoring access to potential spawning habitat through Beaver/dam removal in Enger and Snag Creeks.
• Assessment of size and dynamics of chum salmon populations utilizing the Koidern River.
• Assessment of population levels, health, and dynamics of select freshwater fish populations.

3.2 Donjek River Watershed (not including Kluane River)

The headwaters of the Donjek River originate from the eastern slopes of the St. Elias Mountains within Kluane National Park. The headwaters of the Donjek River are fed by the Kluane, Donjek, Spring, and Steele glaciers. There are no currently occupied communities within the Donjek River watershed, not including that of the Kluane River.

Major tributaries to the Donjek River within the study area include the Klotassin and Nisling Rivers, as well as Wellesley Creek. Thirteen species of fish have been previously documented within the major streams that make up this portion of the Donjek River watershed (see Table 8).

Table 8: Fish species documented in the Donjek watershed

Category	Common Name
Salmon species	Chum salmon, chinook salmon
Freshwater game fish species	Arctic grayling, lake trout, lake whitefish, burbot, northern pike, whitefish general, least cisco
Other fish species	lake chub, slimy sculpin, longnose sucker, round whitefish

3.2.1 Fish Distribution and Habitat

A moderate level of fish diversity was found within this watershed (see Table 9). In general, the level of information available is low, which may be related to the remote nature of many parts of the watershed. Some research has been conducted to identify contemporary chinook spawning and rearing habitat. While these studies have produced some valuable results, a significant understanding of chinook salmon distribution and habitat use has not yet been achieved. Even less information is available in regards to chum salmon spawning in the study area. Some limited research has indicated that chum salmon spawning habitat in the area may be more widespread than is currently understood (Johnston, 1992). Traditional Knowledge has also indicated that this may be the case (Horler, 1994; Penikett, 1993). From a “model of use” perspective, chum salmon spawning habitat can be considered to exist throughout the wide flood plain of a large glacial river such as the Donjek (von Finster pers. comm., 2004). The many remote and inaccessible tributaries and much of the Donjek itself have had, at best, only minimal fish utilization assessments completed. Table 9 lists all streams within the Donjek River watershed with documented species of fish for each. For reference, all streams/species identified in the community workshop have been also listed in a separate column.

Table 9: Streams¹ within the Donjek watershed with documented fish species

<u>Stream Name</u>	<u>Identified by Community in Workshop</u>	<u>Salmon</u>	<u>Freshwater Species</u>
Donjek River	CH	CH, CM	GR, LKC, CCG, RW, BB
Tom Creek			
Klotassin River	CH	CH	BB, CCG
Dip Creek (trib. to Klotassin)		CH	BB, CCG, GR
Nisling River	CH	CH, CM ²	GR
Nansen Creek (trib. to Nisling)			GR
Victoria Creek (trib. to Nisling)			GR, BB
Wellesley Creek	GR	CM ²	GR
Wellesley Lake			LT, NP, WF/LW, RW, BB, LSU, GR
Grayling Creek	GR		GR
Mackinnon Creek			
Hard Luck Creek			
Reed/Murray Creek			
Wolverine Creek			
Ach Creek			
Wade Creek			
Cement Creek			
Steele Creek			
Spring Creek			
Hoge Creek			
Hidden Camp Creek			
Big Horn Creek			

¹Only streams with names appearing on 1:250,000 topographic maps are included.

²Reports of salmon in this location are isolated and not confirmed on a consistent basis.

In general, the level of information regarding habitat utilization of salmon and freshwater fish species within the study area is low. Of the streams for which fish utilization information is available, most indicate a low level of diversity. These include the Klotassin and Nisling Rivers, as well as their tributaries. The Donjek River floodplain habitats and Wellesley Lake are exceptions, showing moderate levels of fish diversity.

The mainstem of the Donjek River is utilized as migration route by chum and chinook salmon bound for the Kluane system. However, chum salmon have also been observed spawning at two different locations downstream of the Kluane River in the Donjek (Johnston, 1992). There is also information indicating the presence of chum salmon in the Nisling River and Wellesley Creek, however these reports have not been confirmed on a consistent basis (Horler, 1994; Penikett, 1993). Spawning chinook salmon have been documented in the Klotassin and Nisling Rivers, with juvenile chinook also identified as rearing in these streams and some tributaries. Thus far, the Nisling has shown the greatest promise as a possibly significant chinook-spawning stream, both in terms of salmon observed/documentated in the system, as well, as available habitat (Eikland 1996, von Finster, 1994 & 1995). However, favourable habitat and chinook spawning/rearing was also identified in the Klotassin River (von Finster, 1994). In general, chinook habitat appears under-utilized in both the Klotassin and Nisling Rivers. It should be noted that in

1891, explorer Charles Willard Hayes referred in his Journal to a First Nation's fish camp on the Nisling River during mid July. He wrote that when he arrived at the camp, "[t]he communal fish-trap had not yet been built, as the salmon were just beginning to ascend; but in a day or two they had it done, and some fine fish were caught" (Hayes, 1891). This is possibly one of the first, if not the first, written documentation of chinook salmon in the Yukon.

3.2.2 Resource Use

Members of the White River First Nation have used this area extensively for generations (Dobrowolsky, 1997; Salmon Workshop, 2004). Some known locations of historic fish camps and settlements have been identified, with certain locations being associated with the harvest of salmon (Hayes, 1891; Salmon Workshop, 2004). However, streams and lakes with significant populations or migrations of freshwater fish have been, and in some cases still are, utilized extensively (Dobrowolsky, 1891; Salmon Workshop 2004).

Guided sports and commercial fisheries were at one time present in Wellesley Lake (FISS, 2004; Penikett, 1993). However, these fisheries no longer exist, and currently only First Nation's subsistence and independent recreational fishers target Wellesley Lake. Table 10 lists some significant historic and current fisheries resource use in the area.

Table 10: Summary of fisheries resource use in the Donjek watershed (excluding Kluane)

Stream	Comments	Data Source
<i>Donjek River</i>	<i>Successful community chinook salmon fishery was undertaken in the mouth of the Donjek River in 1998 with the use of a fish wheel.</i>	<i>Otto, 1999</i>
<i>Klotassin River</i>	<i>"Salmon [chinook] going up there. Did catch some."</i>	<i>Salmon Workshop, 2004</i>
<i>Nisling River</i>	<i>"Salmon [chinook] go up the Nisling. People used to live there, some cabins are there."</i>	<i>Salmon Workshop, 2004</i>
<i>Nisling River</i>	<i>"People used to come here to fish for chinooks and chum."</i>	<i>Horler, 1994</i>
<i>Nisling River</i>	<i>Communal fish-trap constructed/used by two camps, one on either riverbank—mid July, salmon were beginning to run.</i>	<i>Hayes, 1891</i>
<i>Wellesley Lake</i>	<i>Old camp/settlement on this lake. Important year-round fishery location for First Nations from Beaver Creek and Snag, as well as from Coffee Creek. Originally, some people lived there year-round.</i>	<i>Dobrowolsky 1997</i>
<i>Wellesley Lake</i>	<i>At one time was fished heavily by recreational fishers--used to be a lodge on the lake. Also, used to be a commercial lake trout fishery on the lake.</i>	<i>Penikett, 1993</i> <i>FISS, 2004</i>

It is interesting to note the lack of available references to any fishery on the upper Donjek River mainstem, particularly in the vicinity of the Alaska Highway.

3.2.3 Obstructions/Constraints

The primary obstructions to fish passage identified in the Donjek watershed were logjams on the Klotassin and Nisling Rivers (von Finster, 1994, Eikland, 1996). As well, mineral exploration and placer mining may have had (and continue to have) significant detrimental impacts on the water quality in the headwaters of the Nisling River, as well as in a large tributary (Dip Creek) of the Klotassin.

In terms of water quality, extensive mineral exploration from what is known as the “Casino Project” has caused significant disturbance affecting the Dip Creek watershed, which joins the Klotassin River roughly 30km (straight line) downstream from this disturbance. In this case, an oxidizing front in Patent Hill is carrying elevated levels of metals through Casino Creek into Dip Creek (von Finster pers. comm., 2004). Rude Creek, another headwater tributary of Dip Creek, has been extensively affected by placer mining operations (von Finster pers. comm., 2004). The presence of juvenile chinook salmon in Dip Creek has been documented as far as 10 km (straight line) upstream from its confluence with the Klotassin (von Finster, 1994).

Contamination from past hard rock and current/or past placer mining in the headwaters of the Nisling River may also be of some concern. Some contamination of Victoria creek (tributary of Nisling) from tailings pond leakage of the BYG gold mine into Dome Creek (draining into Victoria Creek) was documented in the late 1990s in association with a fish kill at the mouth of Dome Creek (Environmental Protection, 1999). As well, elevated levels of sediment have been identified in Nansen Creek (a tributary of Nisling), and placer mining operations have been active on both Victoria and Nansen Creeks for many years (Environmental Protection, 1999, Osborne, 1995). It should be noted that the upper portions of the Nisling River near Victoria and Nansen Creeks are more than 60 km (straight line) upstream from the nearest documented location (furthest upstream) of spawning chinook salmon. Further, placer operations and extensive mineral exploration activities have also taken place in the upper Klaza River watershed, which joins the Nisling approximately 15 km (straight line) downstream of the most upstream location of spawning chinook salmon documented (von Finster pers. comm., 2004).

The influence of climate change on fish habitat must be considered. Such influence may have current and future effects on a variety of environmental parameters that influence the availability and quality of fish habitat. Such influences may affect surface and subterranean hydrology, alter stream channels and substrates, and change the composition of riparian zone habitats. Such influences are complex with multiple factors directly and indirectly affecting each other and fish habitats.

As well, the question should be considered as to how the natural sediment load of the upper Donjek River (above the Kluane River) may act as a limiting factor in terms of salmon habitat.

3.2.4 Data Gaps and Potential Projects

Significant knowledge gaps have been identified and listed in Table 11.

Table 11: Significant data gaps within the Donjek watershed (as identified by authors)

Limited information regarding the extent of chinook salmon spawning/rearing in the Klotassin River system.
Limited information regarding the extent of chinook salmon spawning/rearing in the Nisling River system.
The extent of impacts on fish habitat from mineral exploration/extraction (including placer) activities in the Klotassin and Nisling watersheds.
The extent of chum spawning in the Donjek River floodplain.
A limited understanding of fresh water fish utilization of streams in the watershed, as well as an understanding in regards to the dynamics and abundance of these populations.

Several ideas related to future work were identified in the community workshop and are listed below in Table 12:

Table 12: Fisheries related ideas for the Donjek watershed identified in community workshop

Gather information that is meaningful for the White River First Nation.
Ensure community involvement, including hiring and training, in projects.
Re-start community fishery for chinook salmon at the mouth of the Donjek. Use/refurbish existing fish-wheel. Possibly combine with scientific research. Involve community and youth as much as possible-----“First Fish” camp, etc... Would like to see this fishery re-established on a more permanent basis.
Study changes in River channels.
Explore options for conducting joint projects with the Kluane First Nation.
Restart salmon in the classroom.
Re-blaze historic trails.

Based on all reviewed information, including ideas/concerns/recommendations identified in the community workshop, some potential projects to be considered for the Donjek River watershed are listed in Table 13 below:

Table 13: Potential fisheries projects for the Donjek watershed (as identified by authors)

Assessment of extent of chinook salmon population utilizing the Klotassin River;
Assessment of extent of chinook salmon population utilizing the Nisling River;
Identify suspected (historical versus present) changes in fish habitat within areas of the Klotassin and/or Nisling watersheds; and, conduct assessment of such changes in relation to possible impacts from mineral exploration/extraction (including placer) activities, as well as those related to climate change;
It should be noted that representatives and members of the White River First Nation have expressed considerable interest in re-starting and maintaining a community fishery for chinook salmon at the mouth of the Donjek River, as was conducted in 1998.

3.3 Kluane River Watershed

The Kluane River drains Kluane Lake and flows northeast for 60-70 km where it joins the Donjek River. Several significant tributaries join the Kluane River directly, including Tincup Creek, Burwash Creek, and the Duke River. Kluane Lake is the largest lake in the Yukon (330,000 hectares) and has considerable importance in terms of fish habitat/utilisation. Kluane Lake is fed by various streams around its perimeter, including the Slims River, Cultus Creek, Gladstone Creek, Talbot Creek, and Brooks Creek. The three currently occupied communities/settlements within the Kluane River watershed are Burwash Landing, Destruction Bay, and Silver City.

Documentation of twelve species of fish within the major streams and lakes of the Kluane River watershed was collected (See Table 14). However, it is likely that more species are present, only that documentation of their presence was not collected in this study.

Table 14: Fish species documented in the Kluane watershed

Category	Common Name
Salmon species	Chum salmon, chinook salmon
Freshwater game fish species	Arctic grayling, lake trout, lake whitefish, burbot, northern pike, round whitefish, inconnu
Other fish species	Slimy sculpin, lake chub, longnose suckers,

It should be noted that emphasis for the collection of information conducted in this study was placed on the upper White and Donjek Rivers. However, a significant amount of documented information regarding the Kluane system was also collected. As such, it should be noted that information collected for the Kluane drainage basin should not be considered a complete account of documented fish habitat utilization, with the possible exception of the salmon species. Due the well-known usage of portions of the Kluane system by chum and chinook salmon, a considerable amount of salmon related information was collected.

3.3.1 Fish Distribution and Habitat

In general, a moderate level of fish diversity was found within streams/lakes for which information was collected (see table 15). Considerable research has been conducted regarding salmon habitat within particular portions of the Kluane drainage system. However, very little research has been pursued on other portions of the watershed known as, or suspected as being, salmon habitat. Further research has been conducted on freshwater fish species associated with Kluane Lake, however, little of this was collected for this project. Table 15 lists all streams within the Donjek River watershed with documented species of fish for each. For reference, all streams/species identified in the community workshop have been also listed in a separate column.

Table 15: Streams¹ within the Kluane watershed with documented fish species

<u>Stream Name</u>	<u>Identified by Community in Workshop</u>	<u>Salmon</u>	<u>Freshwater Species</u>
Kluane River	CH, CM	CH, CM	GR, BB, CCG, LSU, RW, LW, LT, NP, WF
Tom Murray Creek			
Tincup Creek	CH	CH, CM ²	LT (jv), BB, CCG, GR, NP, RW
Tincup Lake		CH ²	LT, GR, LW, RW, BB, CCG
Nuntaea Creek (trib. to Tincup L.)			LT (jv), CCG, GR
Toshingerman Lakes			LT, NP, WF
Swede Johnson Creek		CM ³	GR, LSU, RW, CCG
Quill Creek		CM ³	CCF, LSU, GR
Burwash Creek		CM ³	GR, CCG WF
Duke River		CM ^{2,3}	GR
Kluane Lake	CM	CM, CH ²	LT, LW, GR, BB, IN, LKC, RW, CCG, LSU, NP, WF
Copper Joe Creek			GR, CCG, (LT & LW spawn at mouth in Kluane Lake)
Lewis Creek			
Cluett Creek			
Bock's Creek			
Mines Creek			
Nines Creek			
Congdon Creek			CCG
Williseroft Creek			
Sheep Creek			GR, CCG
Slims River			
Topham Creek			
Silver Creek			
Christmas Creek		CM	GR, CCG, LT, RW, NP, LSU, LW
Inlet Creek			
Cultus Creek		CM ²	GR
Clear Creek			
Gladstone Creek			GR, BB, LT, CCG
Raft Creek			
Talbot Creek			GR
Brooks Creek		CH ²	
Little Creek			
Coon Creek			

¹Only streams with names appearing on 1:250,000 topographic maps are included.

²Reports of salmon in this location are isolated and not confirmed on a consistent basis.

³Chum documented at the mouth of these streams (in Kluane River).

Considerable research has been conducted to identify contemporary chum and chinook spawning and rearing habitat in the Kluane River and Tincup Creek systems. The mainstem of the Kluane River contains well-documented spawning locations for a large run of chum salmon. The lower Kluane River is

also a migratory route for chinook salmon heading to the Tincup system. Further to this, reports of past chinook activity further upstream in the Kluane, particularly at the outlet of Kluane Lake and isolated cases of chinook in Kluane Lake itself, indicate that other (possibly historic) spawning locations of chinook salmon exist in the watershed. It seems, however, that little effort has been made to investigate the extent of other past or present possible chinook spawning areas. The most documented such area, is the outlet of Kluane Lake. The situation is the same in regards to the extent and locations of chum spawning in Kluane Lake and/or its tributaries. It should be noted that the chum salmon populations spawning in Kluane Lake are unique with this regard.

In general, the level of information collected regarding habitat utilization of freshwater fish species within the study area is low. However, as previously mentioned, the information collected in this regard should not be considered a complete account.

3.3.2 Resource Use

Members of the White River First Nation have used this area for generations. Information collected has indicated that streams and lakes with significant populations or migrations of freshwater fish have been, and in some cases still are, utilized extensively. As well past and current participation in the harvest of chum and chinook salmon in the Kluane and Tincup systems has been documented. A guided recreational fishery does currently exist on Tincup Lake. A commercial lake trout fishery continues on Kluane Lake. Other than commercially guided operations, recreational fishing is concentrated primarily along the Alaska Highway corridor, as well as in Kluane Lake. Table 16 lists some significant historic and current fisheries resource use in the area.

Table 16: Summary of fisheries resource use in the Kluane watershed

Stream	Comments	Data Source
<i>Kluane River at "Down Below" (near mouth)</i>	<i>One of the main historic locations for chum harvest, and chinook as well.</i>	<i>Horler, 1994</i>
<i>Tincup Creek</i>	<i>Important historic subsistence salmon fishery.</i>	<i>Dobrowolsky, 1997</i>
<i>Tincup Creek</i>	<i>Historically lots of fish traps for catching chinook were utilized here. Traditional site for hundreds of years. Lots of chinook salmon.</i>	<i>Horler, 1994</i>
<i>Tincup Lake</i>	<i>Fly-In fishing lodge operated by Ruby Range Adventures. Targets Lake Trout and Grayling.</i>	<i>Horler, 1994</i>
<i>Kluane River at Mile 1118</i>	<i>One of the main historic locations for chum harvest.</i>	<i>Horler, 1994</i>
<i>Kluane River at "Salmon Patch," (downstream of Mile 1118)</i>	<i>One of the main historic locations for chum harvest.</i>	<i>Horler, 1994</i>
<i>Kluane Lake</i>	<i>Commercial fishery for lake trout, 7 license holders with a total quota of 2850kg, as of 1994.</i>	<i>Horler, 1994</i>

3.3.3 Obstructions/Constraints

Beaver activity has obstructed chum salmon from reaching their spawning destination in the Glacier Creek slough, a tributary to the Kluane River (von Finster pers. comm. 2004).

The influence of climate change on fish habitat must be considered. Such influence may have current and future effects on a variety of environmental parameters that influence the availability and quality of fish habitat. Such influences may affect surface and subterranean hydrology, alter stream channels and substrates, and change the composition of riparian zone habitats. Such influences are complex with multiple factors directly and indirectly affecting each other and fish habitats.

It should also be noted that the information collected for this study area is not a complete account, and therefore it is likely that further obstructions/constraints exist.

3.3.4 Data Gaps and Potential Projects

Significant knowledge gaps have been identified and listed in Table 17.

Table 17: Significant data gaps within the Kluane watershed (as identified by authors) .

• A limited understanding of spawning destinations and dynamics of chum salmon in Kluane Lake. • A limited understanding of past and/or current chinook spawning destinations other than in the Tincup Creek system, specifically at the outlet of Kluane Lake, and possibly other streams such as Brooks Creek.

Several general ideas related to future work were identified in the community workshop and are listed below in Table 18:

Table 18: Fisheries ideas related to the Kluane watershed identified in community workshop.

• Gather information that is meaningful for the White River First Nation.
• Ensure community involvement, including hiring and training, in projects.
• Explore options for conducting joint projects with the Kluane First Nation.
• Restart salmon in the classroom.
• Re-blaze historic trails.

Based on all reviewed information, including ideas/concerns/recommendations identified in the community workshop, some potential projects to be considered for the Kluane River watershed are listed in Table 19 below:

Table 19: Potential fisheries projects for the Kluane watershed (as identified by authors).

• Assessment of chum spawning areas in Kluane Lake.
• Assessment of past and/or current chinook spawning destinations other than in the Tincup system, particularly in the Kluane River and at the outlet of Kluane Lake.
• Assess feasibility of undertaking marine derived nutrient analysis in locations known or suspected to be historic chinook habitat.

3.4 Upper Tanana River Watershed (upstream of the Tok River)

The upper Tanana River watershed is composed of two headwater river drainages (Chisana and Nabesna) that come together to form the beginning of the Tanana. Several smaller tributaries also join the Tanana River upstream of Tok. The Nabesna and Chisana Rivers are of glacial origin, respectively draining the Nabesna and Chisana glaciers in the Wrangell and Nutzotin mountains. Both streams/glaciers originate in the Wrangell-St. Elias National Park in Alaska. Two tributaries to the Chisana River (Scottie Creek and Mirror Creek) originate in or pass through the Yukon Territory in Canada. A large portion of the upper Tanana watershed is within the Tetlin National Wildlife Refuge, including the lower portions of the Nabesna and Chisana (south of the Alaska Highway) watersheds, as well as that of the Kalutna River on its eastern side, and the upper portion of the Tanana River south of the Alaska Highway. The only settlements within the study area are Tetlin Village, and Northway.

Documentation of fifteen species of fish within the major streams and lakes of the upper Tanana River watershed was collected (See Table 20).

Table 20: Fish species documented in the upper Tanana watershed

Category	Common Name
Salmon species	Chum salmon, chinook salmon, coho salmon
Freshwater game fish species	Arctic grayling, lake trout, lake whitefish, burbot, northern pike, broad whitefish, whitefish general, dolly varden, least cisco
Other fish species	Slimy sculpin, longnose sucker, lake chub, round whitefish

It should be noted that while the central focus of information collected in this study was on the upper White and Donjek Rivers, a certain amount of documented information regarding the upper Tanana system was also collected. Some utilization of the upper Tanana watershed has and does occur by WRFN members. It should therefore be noted that information collected for the upper Tanana River drainage basin should not be considered a complete account of documented fish habitat utilization. Specifically, the Tetlin River and the Chisana River mainstem may have more extensive fish habitat utilization than indicated in this report.

3.4.1 Fish Distribution and Habitat

A moderate to high level of fish diversity was found within streams/lakes for which information was collected (see Table 21). Research has been conducted on specific freshwater species, such as burbot and lake whitefish, within the study area. Further, as much of the area is within the Tetlin National Wildlife Refuge, considerable data has been compiled on the distribution of freshwater species. However, very little research has been pursued regarding the use of the upper Tanana system by salmon. Traditional Knowledge and documented reports of salmon “known to be present” are the only references gathered for this area. Table 21 lists all streams within the Donjek River watershed with documented species of fish for each. For reference, all streams/species identified in the community workshop have been also listed in a separate column.

Table 21: Streams¹ within the upper Tanana watershed with documented fish species

<u>Stream Name</u>	<u>Identified by Community in Workshop</u>	<u>Salmon</u>	<u>Freshwater Species</u>
Tanana River		CH ² , CM, CO ²	BB, LW, DV
Porcupine Creek			
Fish Camp Lakes			LW
Tetlin River	LW, GR, BB		GR, BB, LW
Tetlin Lake			LW
Bitters Creek			GR
Kalutna River			BW, NP, GR
Tlechehan Creek (trib to Kalutna)			BW, NP, GR
Nabesna River			CCG, LSU, LW, RW, GR, NP, BB
Cheslina River			
Jatamund Lake			LT, NP, BB, CS
Beaver Creek			GR, LW, LSU, RW
Chisana River	CM, GR, LW	CH ² , CM, CO ²	GR, LW, BB
Moose Creek			BB
Fish Lake (drains to Moose Cr.)			LW
Takomahto Lake			CS, NP
Tenmile Lake			LW
Gardiner Creek			GR, CCG, RW, LW, NP, LSU
Scottie Creek	CM, WF, GR, NP	CM	LW, BB, GR, LSU, NP, WF
Desper Creek (trib to Scottie)			LW, NP, BB, LSU
Little Scottie Creek (trib to Scottie)			LW
Mirror Creek	LW		GR, LW, LKC, CCG, RW, LSU
American East Wellesley Creek			LW
American East and West Wellesley Creek			CS
Manh Ts'ee (lake)			WF, GR, SU, NP
Sheep Creek		CM	

¹Only streams with names appearing on 1:250,000 topographic maps are included and tributaries of third order streams are only listed if documentation of fish has been collected.

²Reports of salmon in this location are isolated and not confirmed on a consistent basis.

In general, the level of information regarding habitat utilization by salmon within the study area is low. However, the level of information documented in regard to freshwater species is moderate. Species-specific studies on lake whitefish and burbot within the Tetlin National Wildlife Refuge have documented the movements and migrations of said fish. In the case of lake whitefish, several populations have been identified, along with their seasonal habitats and behaviours as adults. Most streams for which fish utilization information is available indicate a moderate to high level of fish diversity. In some cases, such as the Tanana mainstem and the Tetlin River, more information is likely available than was gathered in this project.

Reports of salmon in the upper Tanana system are not well documented. The “Fishery Management Plan” for the Tetlin National Wildlife Refuge states that it “supports a very small run of chum salmon which reaches the refuge in September or October” (Skoog, 1990). It goes on to state that chum salmon “have been reported to occur” in the Chisana River system, including in the Chisana mainstem as well as Scottie Creek and Sheep Creek. The Plan also states that “incidental catches” of chinook, summer chum and coho salmon have been reported from the Chisana River (Skoog, 1990). The only other reference to salmon was from the workshop held in Beaver Creek where participants stated that a few chum were caught for subsistence in the Chisana River, and that salmon used to be found in Scottie Creek (Salmon Workshop, 2004).

3.4.2 Resource Use

Members of the White River First Nation have used this area for generations. Information collected has indicated that streams and lakes with significant populations or migrations of freshwater fish have been, and in some cases still are, utilized extensively. It should be noted that information provided in the Salmon Workshop in Beaver Creek has indicated that the freshwater fish resources in the upper Tanana system have been extremely important to people from the upper White River area and throughout the Tanana River drainage. For example, it was mentioned that during hard times, people would come to Scottie Creek from as far away as Fairbanks to catch whitefish. It was indicated that the area could always be counted on to provide significant quantities of fish (Salmon Workshop, 2004). It should also be noted that the Fisheries Management Plan for the Tetlin National Wildlife Preserve states that historically fish was the staple food of the people in the area. The Plan notes that “[s]everal weirs used to capture fish were located throughout the refuge with the most notable being a stone weir located on Scottie Creek,” again emphasizing the importance of the Scottie Creek system. Currently, such subsistence use of the area continues, however, not necessarily in the same locations or with the same intensity as in the past (Skoog, 1990).

Recreational fishing does occur in the area, however, it is generally limited to areas accessible by road and particular “fly-in” lakes. This includes a limited amount of guided fishing. Grayling, pike, lake trout, and burbot are targeted during the ice-free season, with some ice fishing for pike and burbot during the winter. Rainbow trout have also been stocked into one pothole lake (Hidden Lake) in the area, and recreational fishers target these fish as well (Skoog, 1990). Table 22 lists some significant historic and current fisheries resource use in the area.

Table 22: Summary of fisheries resource use in the Upper Tanana watershed

Stream	Comments	Data Source
<i>Scottie Creek</i>	<i>Historic stone weir site.</i>	<i>Skoog, 2004</i>
<i>Scottie Creek</i>	<i>Very important fishing location for people from all over. People would come from as far away as Fairbanks to fish there during hard times.</i>	<i>Salmon Workshop, 2004</i>
<i>Moose Creek, Fish Creek, Charlieskin Creek, Nabesna River, Chisana River (downstream of Chisana bridge), Fish Lake, Mark Creek, Tenmile Lake, Stuver Creek/Chisana confluence, and Scottie/Desper Creeks.</i>	<i>Fishing areas of Northway residents.</i>	<i>Skoog, 2004</i>
<i>Tlechehen Lake</i>	<i>Active fish camp for whitefish by Tetlin residents.</i>	<i>Skoog, 2004</i>

3.4.3 Obstructions/Constraints

Several current or potential constraints regarding fish populations were identified in the area. Extensive beaver damming was identified as a problem in the Scottie Creek drainage (Salmon Workshop, 2004). As well, several streams along the Alaska Highway have hanging or otherwise problem culverts, thereby reducing the amount of available fish habitat. As well, four placer claims exist in the Cheslina River watershed, however it was not determined if these claims are currently active. Such activity could impact the productivity of fish habitat in the Cheslina and Nabesna Rivers. Other concerns identified were the potential for mineral development in the area (particularly in the headwaters of the Nabesna River), as well as the potential for over harvest of specific fish populations (Skoog, 2004). It should be noted that local residents from Tetlin and Northway have recently expressed concerns that whitefish populations have decreased (Brown, 2003).

The influence of climate change on fish habitat must be considered. Such influence may have current and future effects on a variety of environmental parameters that influence the availability and quality of fish habitat. Such influences may affect surface and subterranean hydrology, alter stream channels and substrates, and change the composition of riparian zone habitats. Such influences are complex with multiple factors directly and indirectly affecting each other and fish habitats.

3.4.4 Data Gaps and Potential Projects

Significant knowledge gaps have been identified and listed in Table 23.

Table 23: Significant data gaps within the upper Tanana watershed (as identified by authors)

• A limited amount of information regarding chum salmon stocks within the upper Tanana system.
• A lack of knowledge regarding the presence/absence of chinook and/or coho salmon in the upper Tanana system.
• A limited amount of freshwater fish population estimates and information on population dynamics.
• A lack of information regarding the impacts of beaver activity, and that of improper stream crossings within the area.

Several general ideas related to future work were identified in the community workshop and are listed below in Table 24:

Table 24: Fisheries ideas related to the upper Tanana watershed identified in community workshop

• Gather information that is meaningful for the White River First Nation.
• Ensure community involvement, including hiring and training, in projects.
• Study Bears gathering salmon (harvest patterns) in Northway.
• Study changes in River channels.
• Beaver and Beaver Dam removal (Scottie Creek).
• Restart salmon in the classroom.
• Re-blaze historic trails.

Based on all reviewed information, including ideas/concerns/recommendations identified in the community workshop, some potential projects to be considered for the upper Tanana River watershed are listed in Table 25 below:

Table 25: Potential fisheries projects for the upper Tanana watershed (as identified by authors)

• Assessment of the presence/absence of chinook and/or coho salmon in the study area.
• Assessment of size and dynamics of chum salmon population utilizing the upper Tanana system, Scottie Creek in particular.
• Assessment of populations and dynamics of key freshwater fish populations.
• Assess feasibility of utilising marine derived nutrient analysis of tree cores in various areas of the upper Tanana and Chisana drainages to establish the historic extent of the presence/absence of salmon.
• Assessment of impacts of beaver activity on fish habitat utilization in Scottie Creek.
• Assessment/mitigation/remediation of hanging culverts along the Alaska Highway that are identified as having an adverse impact on fish habitat utilization.

4.0 CONCLUSIONS

The information documented in this report provides a valuable summary of known fish habitat usage (particularly for salmon) in the WRFN Traditional Territory. The community workshop held in Beaver Creek provided a further opportunity for the community and the WRFN to review and contribute new information to the information compiled herein. Further information was subsequently located, and has also been added. Issues of significance and knowledge gaps were identified for each of the watersheds, as well as potential future projects.

4.1 Significant Issues

Each watershed has some identified issues in terms of obstructions and constraints to fish habitat usage. The issues of most significance¹ have been identified as priorities below:

- An overall decrease in the numbers and extent of salmon returns within the area;
- Limited access to the salmon resource—historically the WRFN utilized salmon (particularly chinook) that were caught at the now abandoned community of Snag and other locations throughout the area. There is considerable interest in the community in once again being able to harvest chinook salmon within the WRFN Traditional Territory;
- Climate change may be affecting availability and quality of fish habitat;
- Extensive log-jams on the Klotassin and Nisling rivers may be obstructing fish passage to important habitats;
- Water quality concerns resulting from mineral exploration/extraction activities in the Klotassin and Nisling Rivers may be impacting fish habitat in these systems;
- Extensive beaver activity may be causing obstructions to fish passage in Scottie Creek;
- Extensive beaver activity may be causing obstructions to fish passage in Snag and Enger Creeks;
- Improper stream crossings along the Alaska Highway may be limiting fish habitat in portions of the upper Tanana River watershed.

4.2 Significant Data Gaps

Each watershed has some identified information gaps in regard to fish habitat usage. Issues requiring further information in areas of most significance have been identified as priorities below:

- The current and historic size and dynamics of the chum salmon populations utilising the Koidern River;
- Present chinook salmon habitat utilization in the upper White River system, upstream of the Donjek;
- Past salmon habitat utilization in the upper White River system—particularly, in the Snag and Sanpete Creek drainages;
- Chinook salmon utilization of the Klotassin Rivers watershed;
- Regarding the significance and extent of chinook spawning habitat in the Nisling River;
- Past/historic significance and extent of chinook returns to the Klotassin and Nisling rivers;
- Extent of impacts on fish habitat from mineral exploration/extraction activities in the Klotassin and Nisling watersheds;

¹ As identified by the authors, refer each watershed for complete listing of all the issues identified at the workshop.

- Extent of chum salmon spawning in the Donjek River mainstem;
- Fish habitat utilization of the White and Donjek watersheds upstream of the Alaska Highway;
- Limited understanding of spawning destinations and dynamics of chum salmon in Kluane Lake;
- Limited understanding of past and/or current chinook spawning destinations other than in the Tincup Creek system, specifically at the outlet of Kluane Lake, and possibly other streams such as Brooks Creek;
- Limited information regarding chum salmon stocks within the upper Tanana system (Scottie Creek may be of particular interest);
- Limited knowledge regarding the presence/absence of chinook and/or coho salmon in the upper Tanana system;
- Limited of information regarding the impacts of beaver activity, and that of improper stream crossings within upper Tanana River system (Scottie Creek may be of particular interest).

4.3 Recommendations

Based upon the above identified issues and information gaps, it is recommended that the WRFN explore opportunities to conduct research and other projects in this regard. A number of potential projects have been identified in each of the watersheds detailed in this report. It is therefore recommended that priority project ideas be pursued to address key information gaps, while at the same time address the needs (particularly the reestablishment of a local salmon fishery) and concerns of the WRFN. In some areas where a common interest exists, it may be valuable to partner with the Kluane First Nation or the Tetlin Tribes in Alaska to conduct such work.

It is further recommended that a central objective of such projects should involve the continued development of interest and capacity within the community of Beaver Creek. As more WRFN members become trained and experienced with regard to fisheries research and management, a sustained local demand and effort will develop regarding the long-term stewardship of fish and their habitats, as well as involvement with the management of the fisheries resource. Involvement in organisations such as the Yukon Salmon Committee, the Yukon River Inter-Tribal Watershed Council and the Aboriginal Fisheries Strategy are valuable methods of staying involved in the fisheries management.

5.0 REFERENCES

Beak Consultants Ltd., 1977(1). Project Report: *Winter fish Investigations of Selected Watercourses in the Yukon Territory*. Prepared for Foothills Pipe Lines (Yukon) Ltd.

Beak Consultants Ltd., 1977(2). Project Report: *A Summer Inventory of the Fishery Resource Along the Proposed Alaska Highway Pipeline in Yukon Territory*. Prepared for Foothills Pipe Lines (Yukon) Ltd.

Beak Consultants Ltd., 1978. Project Report: *A Summary of Fisheries Investigations in Water bodies Within the Influence of the Proposed Alaska Highway Pipeline in Yukon Territory*. Prepared for Foothills Pipe Lines (Yukon) Ltd.

Beak Consultants Ltd., 1979. *Summary of Fisheries Investigation of New Crossing Locations, Alaska Highway Gas Pipeline*. Prepared for Foothills Pipe Lines (Yukon) Ltd.

Blackstone Environmental Services, 2000. *White River First Nation Salmon Harvest and Management Plan*. Prepared for White River FN.

Breaser, Steven W. et al, 1988. U.S. Fish and Wildlife Service. Project Report: *Observations of Movement and Habitat Preferences of Burbot in an Alaskan Glacial River System*.

Brown, Randy J., 2003. U.S. Fish and Wildlife Service. Project Report: *Movement Patterns of Radio-Tagged Adult Humpback Whitefish in the Upper Tanana River Drainage*.

B.Y.G. Natural Resources Inc, 1994. Project Overview: *BYG Mount Nansen Fisheries and Hydrology Trip Report*.

Connor, Mark and Paul Sparling, 1998. White Mountain Environmental Consulting. Project Report: *Report on Preliminary Fisheries Investigations and Training Seminar in the Tincup Drainage, August 1998*.

Connor, Mark and Paul Sparling, 1999. White Mountain Environmental Consulting. Project Report: *Report on Fisheries Investigations in the Tincup Drainage, August 1999*.

Dickson, Gerald, 1996. Kluane First Nation. Project Report: *Tincup Creek Field Trip*.

Dickson, Gerald, 1997. Kluane First Nation. Project Report: *Tincup Field Report*.

Dobrowolsky, Helene, 1997. Project Report: *White River First Nation Land Use and History*. Prepared for White River First Nation.

Eikland, Mark, 1996. Kluane First Nation. Project Report: *Nisling River Field Trip Report*.

Eikland, Mark, 1996. Kluane First Nation. Project Report: *1995 Tincup Creek Salmon Inventory*.

Environment Protection, 1999. Environment Canada. Data Report: *Summer 1998 Field Program Victoria Ck/Nisling River*.

Fernet, D.A. and L.F. Kratt, 1980. Project Report: *Winter Studies of Aquatic Systems Along the Alaska Highway Gas Pipeline in Southern Yukon Territory-Beaver Creek Area. Prepared for Foothills Pipe Lines (Yukon) Ltd.* Prepared for Foothills Pipelines (Yukon) Ltd.

FISS, 2004. Department of Fisheries and Oceans, Fisheries Information Services database. Accessed June & July, 2004.

Foothills Pipe Lines (Yukon) Ltd, 1977. Project Report: *Summary of Spring Watercourse Investigations at Proposed Pipeline Crossing Locations.*

Foothills Pipe Lines (South Yukon) Ltd., 1979. *Environmental Impact Statement for the Alaska Highway Gas Pipeline Project.*

Hayes, Charles Willard, 1891. *Schwatka's last search: New-York ledge expedition through unknown Alaska and British North America: including the Journal of Charles Willard Hayes, 1891.* As published in the New-York ledger magazine as a series of instalments in 1892. University of Alaska Press, 1996.

Horler, Aileen, 1994. Kluane First Nation. *A Review of the Salmon Resource Found Within the Traditional Territory of the Kluane First Nation.*

Johnston, Sandy, 1992. Department of Fisheries and Oceans. Memo: *aerial surveys of the Kluane, Koidern, White and Donjek rivers.*

Kluane First Nation, author and date unknown. *Traditional Knowledge Survey-Results from Salmon Interviews.*

Kratt, L.F. , 1981. Environmental Management Associates. Project Report: *Fisheries Resources of Selected Watercourses Within the Influence of the Alaska Highway Gas Pipeline in the Southern Yukon Territory.* Prepared for Foothills Pipelines (South Yukon) Ltd.

Milligan, P.A. et al, 1984. Department of Fisheries and Oceans. Project Report: *The Distribution and Abundance of Chum Salmon (Oncorhynchus keta) in the Upper Yukon River Basin as Determined by a Radio-Tagging and Spaghetti Tagging Program: 1982-1983.*

Osborne, Clive, 1995. YTG Fish and Wildlife Branch. Memo: *Fish distribution survey of Williams, Victoria, Dome, and Nansen creeks.*

Otto, Darrell K, 1998. Project Report: *Salmon Habitat Assessment in Enger and Beaver Creeks, Yukon Territory.*

Otto, Darrell K., 1997. White River First Nation. Project Report: *Chinook Salmon Habitat Assessment-Beaver and Snag Creeks, Yukon Territory.* Aboriginal Fisheries Strategy.

Otto, Darrell K., 1999. White River First Nation. Project Report: *White River First Nation – Lower Donjek River Chinook Salmon Harvest Feasibility Study.*

Penikett, Lu Johns, 1993. White River First Nation. Project Report: *An Inventory of Resources in the Traditional Territory of White River First Nation: A Basis for Sustainable Resource Management.*

Salmon Workshop, 2004. Notes and recordings of workshop proceedings, held in Beaver Creek, May 27-28, 2004.

Skoog, Ronald O. and Keith M. Schreiner, 1990. *U.S. Fish and Wildlife Service Fisheries Management Plan – Tetlin national Wildlife Refuge.* Prepared for the U.S. Fish and Wildlife Service.

Tobler, Patrick et al, 2001. Environmental Dynamics Inc. Project Report: *2001 Fish and Fish Habitat Inventory and Assessment of the Snag Creek Watershed.* Prepared for Yukon River Panel & White River First Nation.

Tobler, Patrick, 2003. Environmental Dynamics Inc. Project Report: *2002 Upper White River Chinook Reconnaissance.* Prepared for Yukon River Panel & White River First Nation, 2003. Yukon River Panel Project #: CRE-60-02.

Tobler, Patrick, 2003b. Letter Regarding Fish Salvage Works at Burwash Creek on July 13, 2003. Prepared for YTG Transportation Engineering. Prepared by Environmental Dynamics Inc.

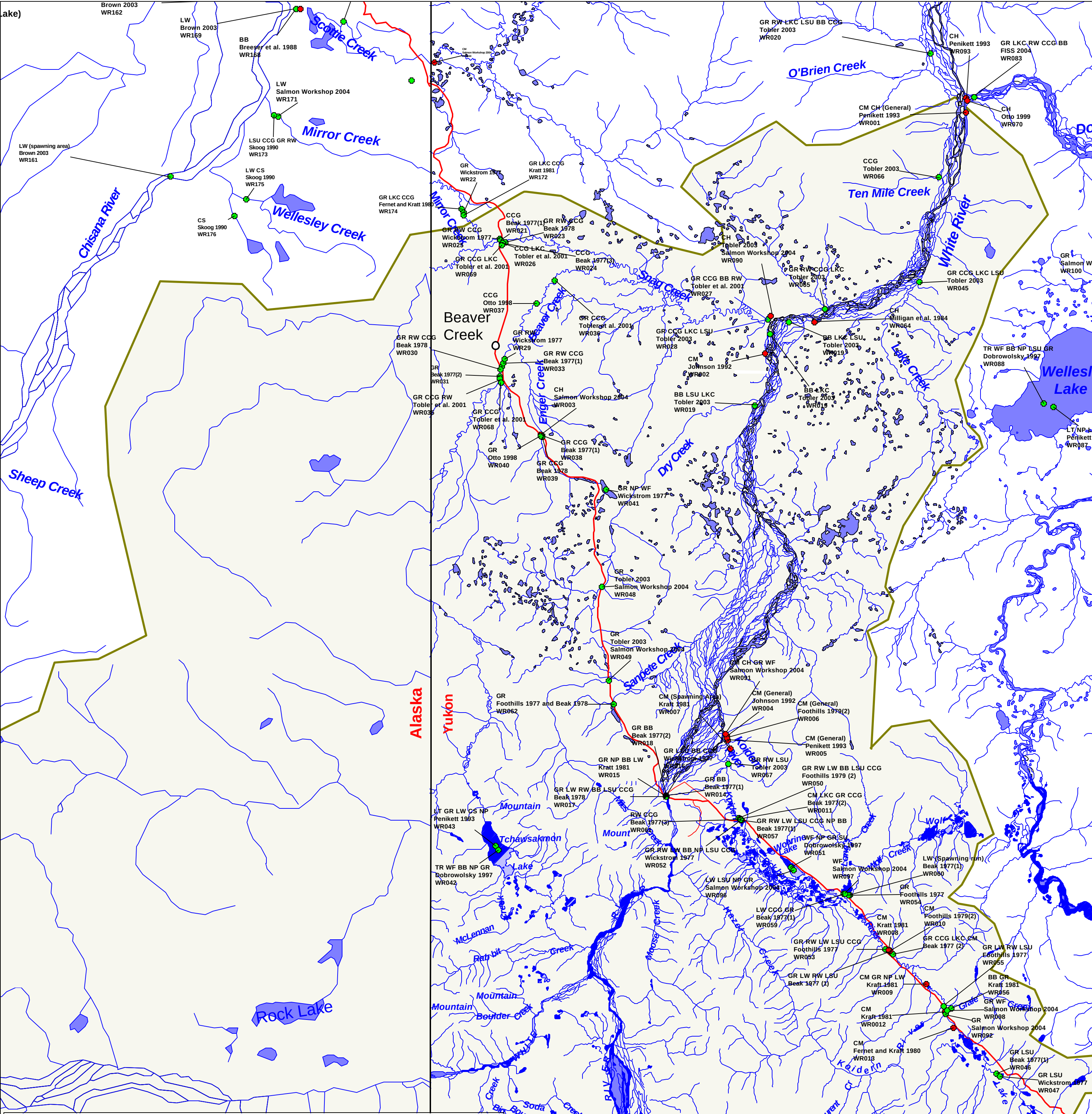
von Finster, Al, 1994. Department of Fisheries and Oceans. Memo: *Overview: waters which may be affected by the Casino Project or by infrastructure associated with it.*

von Finster, Al, 1995. Department of Fisheries and Oceans. Memo: *Overflight of Nisling River.*

von Finster, Al, 2004. Department of Fisheries and Oceans. Personal communication, July 7th & 14th, 2004.

Wilson, Jane & Associates, 2000. Project Report: *Inventory of Chinook Salmon Habitat in the Tincup Creek Drainage, August, 2002.* Prepared for the Kluane First Nation. Yukon River Panel and Aboriginal Fisheries Strategy.

Appendix A:
Fish Distribution Maps



Upper White River Watershed

Scale 1: 250,000

- Lake
- Stream
- Road
- Upper White Watershed
- International Boundary
- Fish Distribution (non-salmon)
- Fish Distribution (salmon)
- Label (includes:
Species code/Reference/ID Number)

FISH SPECIES CODES

- CH chinook salmon
- CM chum salmon
- CO coho salmon
- GR arctic grayling
- CS least cisco
- BW broad whitefish
- LW lake whitefish
- BB burbot
- IN inconnu
- NP northern pike
- LT lake trout
- LSU longnose sucker
- LKC lake chub
- CCG slimy sculpin
- RW round whitefish
- AL arctic lamprey
- WF whitefish (general)

Base maps:

- 1:250,000 for Yukon
- 1:1,000,000 for Alaska

Appendix A provides additional information for each point via ID number. Disclaimer: Locations of points are approximate based on the review of existing literature. This information is presented as a summary and it is recommended that users consult the original literature to verify specific details.



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Appendix A provides additional information for each point via ID number.
Disclaimer: Locations of points are approximate based on the review of existing literature. This information is presented as a summary and it is recommended that users consult the original literature to verify specific details.

Donjek and Kluane Watersheds

Scale 1: 250,000

- Lake
- Stream
- Road
- Donjek Watershed
- Kluane Watershed
- Fish Distribution (non-salmon)
- Fish Distribution (salmon)
- Label (includes:
Species code/Reference/ID Number)

CH
Larry 2004
WR999



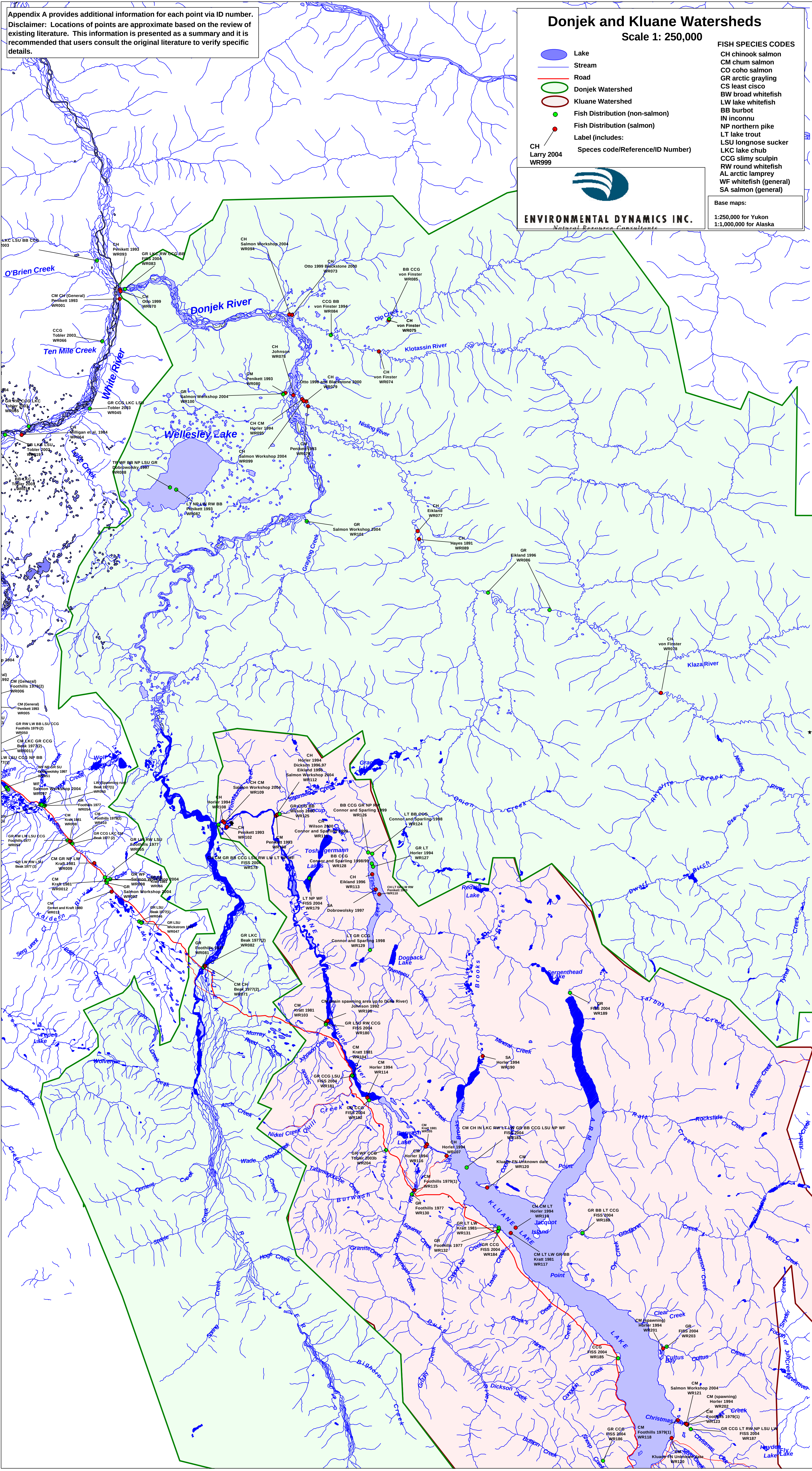
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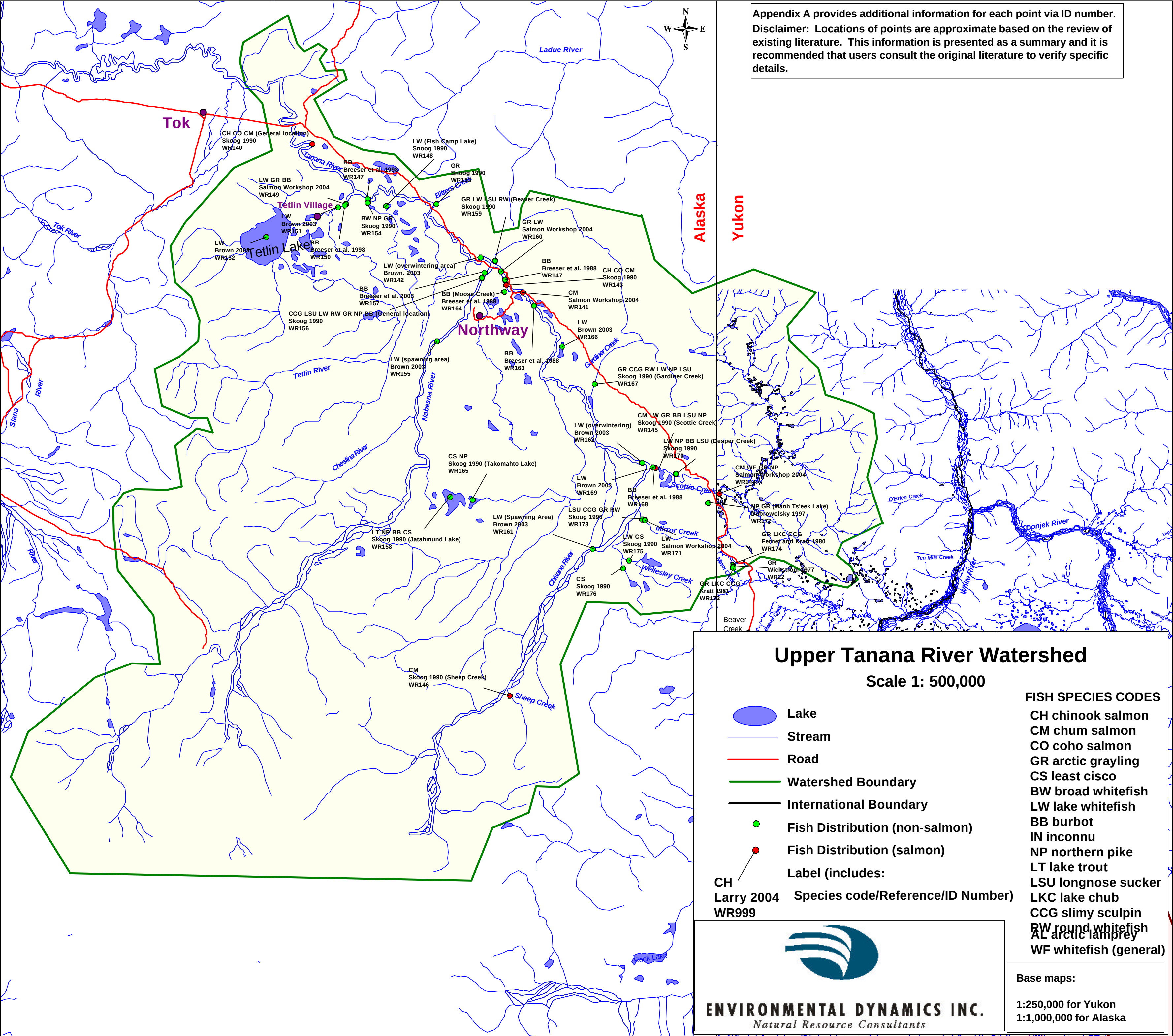
FISH SPECIES CODES

CH chinook salmon
CM chum salmon
CO coho salmon
GR arctic grayling
CS least cisco
BW broad whitefish
LW lake whitefish
BB burbot
IN inconnu
NP northern pike
LT lake trout
LSU longnose sucker
LKC lake chub
CCG slimy sculpin
RW round whitefish
WF whitefish (general)
SA salmon (general)

Base maps:

1:250,000 for Yukon
1:1,000,000 for Alaska





Appendix B:

**Known locations of salmon and freshwater fish species within
the WRFN Traditional Territory and Traditional Area of Use in Alaska**

WHITE RIVER WATERSHED (Upstream of Donjek River)				
Known Locations of Salmon				
ID #	Stream	Location/Comments	Species	Data Source
WR001	White River	Known to be present	CM	Penikett, 1993
WR001	White River	Known to be present	CH	Penikett, 1993
N/A ¹	White River	Use of areas downstream of pipeline/road crossing as a migration route	CM	Kratt, 1981
WR002	White River	3km upstream from Snag Creek, in slough on same side, 2 chum and 2 abandoned redds observed. Incomplete coverage of river.	CM	Johnston, 1992
WR064	White River	Radio tagged chinook located approximately 5km downstream from Snag Creek. Ultimate spawning destination unknown.	CH	Milligan et al, 1984
WR013	Unnamed tributary to Lake Creek	CM eggs observed	CM	Fernet & Kratt, 1980
WR090	Snag Creek	Community of Snag used to exist at the mouth. Chinook were caught in mouth of Snag Creek and the White River—last time in 1942.	CH	Tobler, 2003
WR090	Snag Creek	Used to be catches of chinook near the community of Snag.	CH	Salmon Workshop, 2004
WR003	Enger Creek (Trib. Of Beaver Ck/Snag R.)	Multiple Local Reports in 1995, but no juvenile survival was found in 1996. Also T.K. reports of small numbers of chinook in Enger Creek. Flow of Enger Creek changed after the highway was built.	CH	Salmon Workshop, 2004
WR003	Enger Creek	Chinook sighted migrating upstream in Enger Creek at Alaska Highway crossing.	CH	Otto, 1997
N/A	Sanpete Creek	Documented second hand Traditional Knowledge report of chinook salmon being observed.	CH	Tobler, 2003
WR004	Koidern River	4 adult chum observed in lower 3-4.5 km of River. Historical counts inconsistent, from 1300 in 1984, to 50 in 1987, to 0 in 1988, to 1 in 1990, to 53 in 1991, and 4 in 1992.	CM	Johnston, 1992
WR005	Koidern River	Known to be present	CM	Penikett 1993
WR006	Koidern River	Used by less then 20 chum	CM	Foothills 1979(1)
WR007	Koidern River	KP 79.8, did not migrate past this point that year but a spawning area was noted 8 km downstream. Most use documented downstream of crossing.	CM	Kratt 1981

Continued...

¹ N/A—Information provided is general in nature and/or the location could otherwise not be mapped.

WHITE RIVER WATERSHED (Upstream of Donjek River)				
Known Locations of Salmon Continued...				
ID #	Stream	Location/Comments	Species	Data Source
WR008	Koidern River	KP 98.7, occasionally chum migrate through this area to spawn in upper reaches of river.	CM	Kratt 1981
WR009	Koidern River	KP 108.8 one CM fry and viable eggs were collected 3 km downstream.	CM	Kratt 1981
N/A	Koidern River	Upper reaches, spawning and migration	CM	Kratt 1981
WR010	Koidern River	KPM101.4 near crossing site	CM	Foothills 1979(2)
WR011	Koidern River	Mile 49.2 downstream	CM	Beak 1977(2)
N/A	Koidern River	Pipeline crossing	CM	Beak 1978
WR012	Koidern River	Redds at mouth of Grafe Creek—KP 109.5 crossing.	CM	Beak in Kratt, 1981
N/A	Koidern River	“Chinook come here to spawn, but not very many. Chum also come here.” (T.K.)	CH, CM	Horler, 1994
WR091	Koidern River	Lots of chum are or used to be found there, chinook used to go there—oral reports from 1950-55.	CH, CM	Salmon Workshop, 2004
WHITE RIVER WATERSHED (Upstream of Donjek River)				
Fresh Water Fish Locations				
ID #	Stream	Location/Comments	Species	Data Source
WR014	White River	At crossing	GR (jv), BB (jv) (July 30)	Beak 1977(1)
WR015	White River	At KP 72.8 pipeline crossing	GR (jv), NP, BB, LW	Kratt 1981
WR016	White River	Mile 44.3, large numbers of jv GR using side channels and backwaters of river	GR, LSU, BB, CCG	Foothills, 1977
N/A	White River	Mile 43.5-44.5	GR	Foothills, 1977
WR017	White River	Mile 44.3	GR, LW, RW, BB, LSU, CCG	Beak 1978
WR018	White River	Mile 44.25	GR, BB	Beak 1977(2)
WR019	White River	Various locations from mouth of Donjek to Alaska Hwy.	BB, LKC, LSU	Tobler, 2003
WR020	O'Brien Creek	Just across and downstream from mouth of Donjek River. Within flood plain of the White River.	GR, RW, BB, CCG, LKC, LSU	Tobler, 2003
WR066	Ten Mile Creek	Sampling near mouth	CCG	Tobler, 2003
WR065	Unnamed Tributary to White River.	Approx. 5km below Snag Creek, on same side.	GR, RW, CCG, LKC	Tobler, 2003
WR021	Snag Creek	At crossing	CCG (August 2)	Beak 1977(1)
WR022	Snag Creek	Mile 9.0	GR	Foothills, 1977
WR023	Snag Creek	Mile 9.0	GR, RW, CCG	Beak 1978
WR024	Snag Creek	Mile 8.9	CCG	Beak 1977(3)
WR025	Snag Creek	Mile 9.0	RW, GR, CCG	Foothills, 1977
WR026	Snag Creek	At highway crossing	CCG, LKC	Tobler et al, 2001
WR027	Snag Creek	Lower reaches, 1km from mouth	GR, CCG, BB, RW	Tobler et al, 2001
WR028	Snag Creek	Lower reaches, 1km from mouth	GR, CCG, LSU, LKC	Tobler, 2003

Continued...

WHITE RIVER WATERSHED (Upstream of Donjek River)				
Fresh Water Fish Locations Continued...				
ID #	Stream	Location/Comments	Species	Data Source
WR069	Unnamed Tributary to Snag Creek.	Small drainage adjacent to Old Alaska Highway Location	GR, CCG, LKC	Tobler et al, 2001
WR029	Beaver Creek (trib. to Snag)	Mile 14.8 near crossing	GR, RW	Foothills, 1977
WR030	Beaver Creek	Mile 14.8	GR, RW, CCG	Beak 1978
WR031	Beaver Creek	Mile 15	GR	Beak 1977(2)
WR032	Beaver Creek	PKM 23.8 near crossing	GR, RW, CCG	Beak 1979(2)
WR033	Beaver Creek	At crossing	GR, RW (jv), CCG	Beak 1977(1)
WR034	Beaver Creek	Used for spawning, nursery, & rearing habitat.	GR	Beak 1979(2)
WR035	Beaver Creek	Above Highway crossing	GR, CCG, RW	Tobler et al, 2001
WR036	Beaver Creek	Roughly ½ way from highway to confluence w/ Snag Creek	GR, CCG	Tobler et al, 2001
WR037	Beaver Creek	7km downstream from Alaska Hwy.	CCG	Otto, 1998
WR68	Unnamed Tributary to Beaver Creek	Small stream draining wetland, lots of juvenile grayling	GR	Tobler et al, 2001
WR038	Enger Creek (trib to Beaver) (mile 15.3)	At crossing	GR(fry), CCG (August 1)	Beak 1977(1)
WR039	Enger Creek	Mile 15.3, 18.7, 23.8	GR, CCG	Beak 1978
WR040	Enger Creek	At Alaska Hwy Crossing	GR	Otto, 1998
WR041	Enger Lakes	Mile 23.5	GR, NP, WF	Foothills, 1977
WR042	Tchawsahmon Lake		LT, WF, BB, NP, GR	Dobrowolsky 1997
WR043	Tchawsahmon Lake	Based on traditional knowledge sources	LT, GR, HW/LW, CS (least), NP	Penikett 1993
WR044	Fish Hole Lake	Important fishery for Beaver Creek people	GR, NP	Dobrowolsky 1997
WR045	Lake Creek (trib. to White River)	Lower Reaches	GR, CCG, LSU, LKC	Tobler, 2003
WR048	Dry Creek	In fall & spring, stream is dry in the summer	GR	Salmon Workshop, 2004
WR048	Dry Creek		GR	Tobler, 2003
WR049	Sanpete Creek		GR	Tobler, 2003
WR049	Sanpete Creek	Just grayling there	GR	Salmon Workshop, 2004
WR057	Koidern River	At Crossing (mile 48.9)	GR, LW, RW, LSU, CCG (July 30)	Beak 1977(1)
WR058	Koidern River	At Crossing (mile 63)	GR, CCG, LW	Beak 1977(1)
WR009	Koidern River	KP 108.8 in vicinity of pipeline crossing	GR, NP, LW	Kratt 1981
WR050	Koidern River	PKM 101.4 near crossing	GR, LW, RW, BB, LSU, CCG	Foothills 1979(2)
WR052	Koidern River	Mile 48.9 near crossing—grayling spawning grounds identified 800m below	GR, RW, LW, BB, NP, LSU, CCG	Foothills, 1977
WR053	Koidern River	Mile 63.0 near crossing	GR, RW, LW, CCG, LSU	Foothills 1977
WR011	Koidern River	Mile 49, 2 km downstream	LKC, GR, CCG	Beak 1977(2)

Continued...

WHITE RIVER WATERSHED (Upstream of Donjek River)				
Fresh Water Fish Locations Continued...				
ID #	Stream	Location/Comments	Species	Data Source
WR061	Koidern River	Mile 48.2, winter sampling	RW, CCG	Beak 1977(3)
WR055	Koidern River	Mile 68.1, stream incorrectly identified as Edith Creek, map shows it as Koidern.	GR, LW, RW, LSU	Foothills, 1977
N/A	Koidern River	KP 99.8, march	CCG	Fernet & Kratt, 1980
WR091	Koidern River		WF, GR	Salmon Workshop, 2004
WR067	Koidern River	Approx. 5 km downstream from highway crossing, abundant adult fish observed and captured.	GR, RW, LSU	Tobler, 2003
WR051	Pickhandle Lake		WF, NP, GR, SU	Dobrowolsky 1997
WR096	Pickhandle Lake	Lots of Whitefish	LW, LSU, NP, GR	Salmon Workshop, 2004
WR059	Wolf Creek (trib. to Koidern)	At Crossing	LW (fry), CCG, GR	Beak 1977(1)
WR097	Wolf Creek	Lots of Whitefish (likely Lake Whitefish)	WF/LW	Salmon Workshop, 2004
WR060	Wolf Creek	Local residents report a spawning run, sampling indicates the area is used as a nursery/rearing area.	LW	Beak 1977(1)
WR054	Wolf Creek	Mile 59.0 near crossing	GR	Foothills, 1977
WR056	Grafe Creek (trib. to Koidern)	KP 109.5 crossing	BB (jv), GR (jv)	Kratt 1981
WR098	Grafe Creek	Grayling and Whitefish found there. (likely Lake Whitefish).	GR, WF/LW	Salmon Workshop, 2004
WR062	Unnamed Creek (trib. to Koidern)	Mile 37.8 appeared to be migrating upstream into a lake	GR	Foothills, 1977
WR046	Lake Creek (Trib. to Koidern R.)	Mile 68.5 near crossing	GR, LSU (spawning areas)	Foothills, 1977
WR047	Lake Creek (Trib. to Koidern R.)	At crossing	GR (fry), LSU (fry)	Beak 1977(1)
N/A	Lake Creek (Trib. to Koidern R.)		GR	Salmon Workshop, 2004
WR062	Unnamed Creek	Mile 37.8	GR	Beak 1978
N/A	Salt Lake	Lake NW of Snag	NP (lots)	Dobrowolsky 1997

DONJEK RIVER WATERSHED (Except Kluane Drainage)				
Known Locations of Salmon				
ID #	Stream	Location/Comments	Species	Data Source
WR070	Donjek River	Fish Wheel Harvest near mouth in 1998	CH	Otto, 1999
WR093	Donjek River	Known to be present	CH	Penikett 1993
WR071	Donjek River	Mile 80	CM, CH	Beak 1977(2)
WR072	Donjek River	2 spawning sites located, one between Wellesley Creek and Lynx City, & one upstream of the Nisling River. Incomplete coverage of river.	CM	Johnston, 1992
WR073	Klotassin River	Adults and juveniles present	CH	Otto/Blackstone, 1998/2000
WR094	Klotassin River	Salmon went up, some were caught.	CH	Salmon Workshop, 2004
WR074	Klotassin River	Adults and juveniles confirmed in lower 15km (straight line)	CH	von Finster, 1994
WR075	Dip Creek (trib. to Klotassin)	Juveniles found in lower 10km of Dip Creek	CH	von Finster, 1994
WR076	Nisling River	Known to be present	CM	Penikett 1993
WR077	Nisling River	Observed small numbers of adults in lower 30km (straight line) of the River.	CH	Eikland, 1996
WR078	Nisling River	Observed adults and spawning redds in lower 70km (straight line) of the River.	CH	von Finster, 1995
WR079	Nisling River	Adults and juveniles present	CH	Otto/Blackstone, 1998/2000
WR095	Nisling River	“People used to come here to fish for Chinooks and chum.” “King Salmon come up the Nisling” (T.K)	CH, CM	Horler, 1994
WR099	Nisling River	Chinook were found there. People used to live there, cabins are/were there.	CH	Salmon Workshop, 2004
WR089	Nisling River	“communal fish-trap” built/operated by F.N. camps on both river banks. Early July, salmon were just starting to run.	CH	Hayes, 1891
WR080	Wellesley Creek	Known to be present	CM	Penikett 1993

DONJEK RIVER WATERSHED (Except Kluane Drainage)				
Fresh Water Fish Locations				
ID #	Stream	Location/Comments	Species	Data Source
WR081	Donjek River	Mile 81.0 near crossing	GR	Foothills, 1977
WR082	Donjek River	Mile 80	GR, LKC,	Beak 1977(2)
WR083	Donjek River	General- (river as a whole)	GR, LKC, RW, CCG, BB	FISS, 2004
WR084	Klotassin River	Mouth of Dip Creek	CCG, BB	von Finster, 1994
WR085	Dip Creek (trib. to Klotassin)	10km (straight line) upstream from mouth. Only s. sculpins caught 27 km (straight line) upstream from mouth, but dead grayling also observed.	CCG, BB CCG, GR	von Finster, 1994
WR086	Nisling	16 & 26 km below Klaza River	GR	Eikland, 1996
N/A	Nansen Creek (trib. to Nisling)	1 km upstream from Webber Ck.	GR	BYG, 1994
N/A	Victoria Creek (trib. to Nisling)		GR, BB	Osborne, 1995
N/A	Victoria Creek (trib. to Nisling)		GR	BYG, 1994
WR100	Wellesley Creek	Grayling were found there.	GR	Salmon Workshop, 2004
WR087	Wellesley Lake	Based on traditional knowledge sources	LT, NP, LW, RW, BB	Penikett 1993
WR088	Wellesley Lake		LT, WF, BB, NP, LSU, GR	Dobrowolsky 1997
WR101	Grayling Creek	An old village was there before. (T.K.)	GR	Salmon Workshop, 2004

KLUANE RIVER WATERSHED				
Known Locations of Salmon				
ID #	Stream	Location/Comments	Species	Data Source
WR 178	Kluane River	General—stream summary	CH, CM	FISS 2004
WR102	Kluane River	Known to be present	CH	Penikett 1993
WR103	Kluane River	KP 150.6 crossing, spawned at mouth of Swede Johnson Creek	CM	Kratt 1981
WR104	Kluane River	KP 160.2, spawned at mouth of Quill Creek	CM	Kratt 1981
WR105	Kluane River	KP 160.2, spawned at mouth of Duke River	CM	Kratt 1981
WR106	Kluane River	Spawning chum surveyed from Duke River to below Swede Johnson Creek. (main area of chum spawning)	CM	Johnston, 1992
N/A	Kluane River	Chum were caught in large numbers at various locations (Down Below, Swede Johnson, Quill Creek, Mile 1118, Salmon Patch, Mile 1114-5) from the mouth of the river all the way to the Duke River and Kluane Lake. (T.K.)	CM	Kluane F.N., Date Unknown
N/A	Kluane River	Chum salmon were fished, primarily for dog food, at various locations along the entire length of the Kluane River “Lots of chum salmon come here to spawn in the fall. The can be seen packed into the mouth of Swede Johnson Creek although not in the creek itself.” “They spawn all along the river in the sloughs.” (T.K.)	CM	Horler, 1994
WR107	Kluane River	Lots of chinook used to spawn in outlet of Kluane Lake and streams nearby, but not any more. (T.K.)	CH	Horler, 1994
WR108	Kluane River	“King Salmon, we used to catch them Down Below [near mouth of Kluane River] in Mid-August.”	CH	Horler, 1994
WR109	Kluane River	Chum were caught there, also chinook have been caught there. Report of fishing for chinook in Kluane River, gaffed 16 in one evening---unsure of location.	CM, CH	Salmon Workshop, 2004
WR190	Tincup Creek	Known to be present	CM	Penikett 1993
WR111	Tincup Creek	Adults observed and Juveniles captured	CH	Wilson, 2000
WR111	Tincup Creek	Adults observed and Juveniles captured	CH	Connor & Sparling, 1999
WR112	Tincup Creek	Site of significant historic gathering/fishing location—several accounts of using fish traps to catch chinook (T.K.)	CH	Horler, 1994
WR112	Tincup Creek	Spawning salmon observed	CH	Dickson, 1997

Continued...

KLUANE RIVER WATERSHED				
Known Locations of Salmon Continued...				
ID #	Stream	Location/Comments	Species	Data Source
WR112	Tincup Creek	Spawning salmon counted	CH	Dickson, 1996
WR112	Tincup Creek	Spawning salmon counted—historically supported much larger run (only 119 counted)	CH	Eikland, 1996
WR112	Tincup Creek		CH	Salmon Workshop, 2004
WR110	Tincup Lake	Known to be present	CH	Penikett 1993
WR113	Tincup Lake	One chinook observed in lake	CH	Eikland, 1996
WR114	Burwash Creek	Chum are found there	CM	Horler, 1994
WR115	Duke River	Near proposed crossing	CM (carcass)	Foothills 1979(1)
WR116	Duke River	Chum are found there	CM	Horler, 1994
WR117	Kluane Lake	KP 220.5-226.2	CM	Kratt, 1981
WR118	Kluane Lake	As far south as Christmas Bay	CM	Foothills 1979(1)
WR119	Kluane Lake	-Chum are found all over the lake, all the way to Slims River. -A large net full of chum was once pulled from Dutch Harbour. -Chum spawn in Cultus and Christmas Creeks.	CM	Horler, 1994
WR119	Kluane Lake	Isolated catches of chinook have occurred in the past, including one in Brooks Arm, and one “way up the lake.”	CH	Horler, 1994
WR120	Kluane Lake	Locations identified where people fished for chum salmon: -Point-Burwash Landing; -across Kluane Lake from Burwash; -Dutchman Point; -Silver Creek. (T.K.)	CM	Kluane F.N., Date Unknown.
WR121	Kluane Lake	Chum in Kluane Lake, spawning in Lake/Christmas Creek, possible other spawning locations on the western shore of the lake.	CM	Salmon Workshop, 2004
N/A	Kluane Lake	chum spawning area at Mary’s Point	CM	Foothills 1979(1)
WR190	Brooks Creek	Ethnographic study from 1964 describes the presence of salmon in Brooks and Tincup Creeks. Most likely a reference to chinook.	CH	Johnson & Raup in Horler, 1994
WR123	Christmas Creek near mouth	Chum spawning	CM	Foothills 1979(1)
WR202	Christmas Creek	Chum spawning	CM	Horler, 1994
WR201	Cultus Creek	Chum spawning	CM	Horler, 1994

KLUANE RIVER WATERSHED				
Fresh Water Fish Locations				
ID #	Stream	Location/Comments	Species	Data Source
WR178	Kluane River	General—stream summary	GR, BB, CCG, LSU, RW, LW, LT, NP, WF	FISS 2004
WR124	Tincup Creek	Juvenile sampling from lake outlet, to 2 km downstream Unusually high densities of sculpins found in lake outlet.	LT, BB, CCG	Connor & Sparling, 1998
WR125	Tincup Creek	Juvenile sampling throughout Tincup Cr.	GR, CCG, BB	Wilson, 2000
WR126	Tincup Creek	Juvenile sampling & snorkelling in upper portions of the creek	BB, CCG, GR, NP, RW	Connor & Sparling, 1999
WR110	Tincup Lake	Based on traditional knowledge sources	LT, GR, LW, RW	Penikett 1993
WR127	Tincup Lake	Fishing Lodge, clients targeting mainly grayling and lake trout	GR, LT	Horler, 1994
WR128	Tincup Lake	Juvenile sampling in T. Lake	BB, CCG	Connor & Sparling, 1998, & 1999
WR129	Nuntaea Creek (trib. to Tincup L.)	Juvenile sampling	LT, CCG, GR	Connor & Sparling, 1998
WR179	Toshingerman Lakes	General Summary	LT, NP, WF	FISS, 2004
WR180	Swede Johnson Creek	General Summary	GR, LSU, RW, CCG	FISS, 2004
WR181	Quill Creek	General Summary	CCG, LSU, GR	FISS, 2004
WR182	Burwash Creek	General Summary	GR, CCG	FISS, 2004
WR130	Duke River	mile 111.0 near crossing	GR	Foothills, 1977
WR183	Kluane Lake	General—Lake Summary	IN, LKC, RW, LT, LW, GR, BB, CCG, LSU, NP, WF	FISS, 2004
WR117	Kluane Lake	KP 220.5-226.2	LT, LW, GR, BB	Kratt 1981
WR119	Kluane Lake	Commercial fishery for Lake Trout, 2,850 kg quota	LT	Horler, 1994
WR184	Copper Joe Creek	General Summary	GR, CCG	FISS, 2004
WR131	Copper Joe Creek	KP 191.1 crossing	GR, LT and LW spawn at mouth in Kluane Lake	Kratt 1981
WR132	Copper Joe Creek	A mouth of creek	GR	Foothills, 1977
WR185	Congdon Creek	General Summary	CCG	FISS, 2004
WR186	Sheep Creek	General Summary	GR, CCG	FISS, 2004
WR187	Christmas Creek	General Summary	GR, CCG, LT, RW, NP, LSU, LW	FISS, 2004
WR203	Cultus Creek	General Summary	GR	FISS, 2004
WR188	Gladstone Creek	General Summary	GR, BB, LT, CCG	FISS, 2004
WR189	Talbot Creek	General Summary	GR	FISS, 2004
WR204	Burwash Creek	At highway crossing. WF displayed characteristics of both round and pygmy whitefish.	GR, CCG, WF	Tobler 2003b

UPPER TANANA WATERSHED (upstream of Tok River)				
Known Locations of Salmon				
ID #	Stream	Location/Comments	Species	Data Source
WR140	Tanana River	Similar fish distribution/utilization as Chisana & Nabesna Rivers.	CH CM CO	Skoog 1990
WR141	Chisana River	Northway, a few caught, but not many	CM	Salmon Workshop, 2004
WR143	Chisana River	Have been reported to occur	CM	Skoog 1990
WR143	Chisana River	Incidental catches of chinook, coho, and summer chum have been reported	CH, CO, CM	Skoog, 1990
WR144	Scottie Creek	Used to be chum in Scottie Cr, just above border (1930s).	CM	Salmon Workshop, 2004
WR145	Scottie Creek	Have been reported to occur	CM	Skoog 1990
WR146	Sheep Creek	Spawning reported	CM	Skoog 1990
UPPER TANANA WATERSHED (upstream of Tok River)				
Fresh Water Fish Locations				
ID #	Stream	Location/Comments	Species	Data Source
WR147	Tanana River	Burbot radio-tagged at the mouth of the Kalutna River moved upstream to Chisana and Nabesna Rivers. Some burbot radio-tagged in the Chisana River moved downstream in Tanana River.	BB	Breaser et al, 1988
WR142	Tanana River	Extensive over-wintering location downstream of Nabesna/Chisana confluence.	LW	Brown, 2003
N/A	Tanana River*	*Dolly varden have been found in some streams in the area, however specific stream locations have not been named.	DV	Skoog, 1990
WR148	Fish Camp Lakes	General	LW (Humpback)	Skoog 1990
WR149	Tetlin River		LW, GR, BB	Salmon Workshop, 2004
WR150	Tetlin River	Radio tagged burbot relocated here—possible spawning area.	BB	Breaser et al, 1988
WR151	Tetlin River	Migration route for lake whitefish.	LW	Brown, 2003
WR152	Tetlin Lake	Over wintering habitat for lake whitefish	LW	Brown, 2003
WR153	Bitters Creek	General	GR	Skoog 1990
WR154	Kalutna River	Upper portions appear to be spawning area for grayling, and pike caught just above Tlechehn Creek. Appears to be a migration route for broad whitefish headed for Tlechehn Creek.	BW, NP, GR	Skoog, 1990
N/A	Tlechehn Creek (trib to Kalutna)	Significant amount of broad whitefish caught in sampling.	BW	Skoog, 1990
WR155	Nabesna River	Spawning area 15 - 30 km upstream from its mouth	LW (humpback)	Brown, 2003
WR156	Nabesna River	General	CCG LSU LW (Humpback) RW GR NP BB	Skoog, 1990

Continued...

UPPER TANANA WATERSHED (upstream of Tok River)				
Fresh Water Fish Locations Continued...				
ID #	Stream	Location/Comments	Species	Data Source
WR157	Nabesna River	Radio tagged burbot relocated here.	BB	Breaser et al, 1988
WR158	Jatahmund Lake	Only lake with Lake Trout found in Tetlin refuge	LT NP BB CS	Skoog 1990
WR159	Beaver Creek	General	GR LW (Humpback) LSU RW	Skoog 1990
WR160	Chisana River		GR, LW	Salmon Workshop, 2004
WR161	Chisana River	Spawning area 80 - 100 km upstream from mouth	LW	Brown, 2003
WR162	Chisana River	Extensive over-wintering areas immediately downstream of Scottie Creek and half way between Scottie Creek and Nabesna River.	LW	Brown, 2003
WR143	Chisana River	General	BB	Skoog 1990
WR163	Chisana River	Radio tagged burbot travelled upstream from near Moose and Mark Creeks to possible spawning area at Gardiner Creek Flats.	BB	Breaser et al, 1988
WR164	Moose Creek	Radio tagged burbot relocated here.	BB	Breaser et al, 1988
N/A	Fish Lake (drains to Moose)	Spring/early summer habitat.	LW	Brown, 2003
WR165	Takomahto Lake	General	CS NP	Skoog 1990
WR166	Tenmile Lake	Spring/early summer habitat.	LW	Brown, 2003
WR167	Gardiner Creek	General	GR CCG RW LW (Humpback) NP LSU	Skoog 1990
WR144	Scottie Creek	Just above Border	WF, GR, NP	Salmon Workshop, 2004
WR145	Scottie Creek	General	LW (Humpback) GR BB LSU NP	Skoog 1990
WR168	Scottie Creek	Radio tagged burbot relocated here.	BB	Breaser et al, 1988
WR169	Scottie Creek	Spring/summer/over-winter habitat	LW	Brown, 2003
WR170	Desper Creek (trib to Scottie)	Rearing and possibly spawning stream for whitefish and pike, as well as rearing for burbot and suckers.	LW, NP, BB, LSU	Skoog, 1990
N/A	Little Scottie Creek (trib to Scottie)	General	LW (Humpback)	Skoog 1990
WR171	Mirror Creek	Whitefish come up in the fall time (spawning).	LW	Salmon Workshop, 2004
WR172	Mirror Creek	At pipeline crossing	GR, LKC, CCG	Kratt 1981
WR173	Mirror Creek	General	RW GR CCG LSU	Skoog 1990
N/A	Mirror Creek	substantial population	GR	Foothills, 1977
WR174	Mirror Creek	KP 7.2, march	GR, LKC, CCG	Fernet & Kratt, 1980

Continued...

UPPER TANANA WATERSHED (upstream of Tok River)				
Fresh Water Fish Locations Continued...				
ID #	Stream	Location/Comments	Species	Data Source
WR175	American East Wellesley Creek	General	LW (Humpback)	Skoog 1990
WR176	American west Wellesley	General	CS	Skoog 1990
WR177	Manh Ts'eeek	Important fishery for Tanana people	WF, GR, SU, NP (all abundant)	Dobrowolsky 1997

Appendix C:

Salmon Information Workshop Participants, Beaver Creek 2004.

Salmon Information Workshop Participants, Beaver Creek 2004.

Name	Organisation/Affiliation
Agnes Winzer	Local Resident
Chief Angela Davis	WRFN—Chief
Brad Wilson	Habitat Steward—Yukon Fish and Wildlife Management Board
Carl Sydney	Yukon Salmon Committee
Charlie Eikland Jr.	Local Resident (day 2 only)
Connie Friend	Tetlin National Wildlife Refuge (Tok)
Connie Larochelle	WRFN—Land Claims
David Johnny	Local Resident
Ellen Klemmit	Tetlin Elder (Tok)
Isaac Anderton	Environmental Dynamics Inc.
Joe Johnson	Yukon Salmon Committee
Joleene Murphy	WRFN—Councillor
Kaz Kuba	WRFN—Land Claims
Lorelie Smith	Yukon Salmon Committee
Mary Blair	Local Elder
Patrick Johnny	WRFN—Land Claims
Patrick Milligan	Fisheries and Oceans Canada
Patrick Tobler	Environmental Dynamics Inc.
Sid Vandermeer Sr.	Local Resident (open house only)
Stan Peters	Local Resident (open house and day 2 only)